一、项目整体情况介绍

1.1项目整体信息概况

项目信息

项目编号	2092138333112520706
项目名称	测试阿萨德发
所属项目组	分析测试
项目描述	阿道夫
创建时间	2026-08-25 14:33:43
完成时间	2026-08-25 22:40:01

1.2分析流程

测序数据 > 数据质控与比对 > 表达量分析 > 差异表达分析 > 功能富集分析 > 蛋白互作网络分析 > 可变剪切分析 > 融合基因分析

二、数据质控分析

对原始测序数据进行质量评估，去除接头污染和低质量数据，并对质控前后数据进行多维统计。

2.1数据整理表

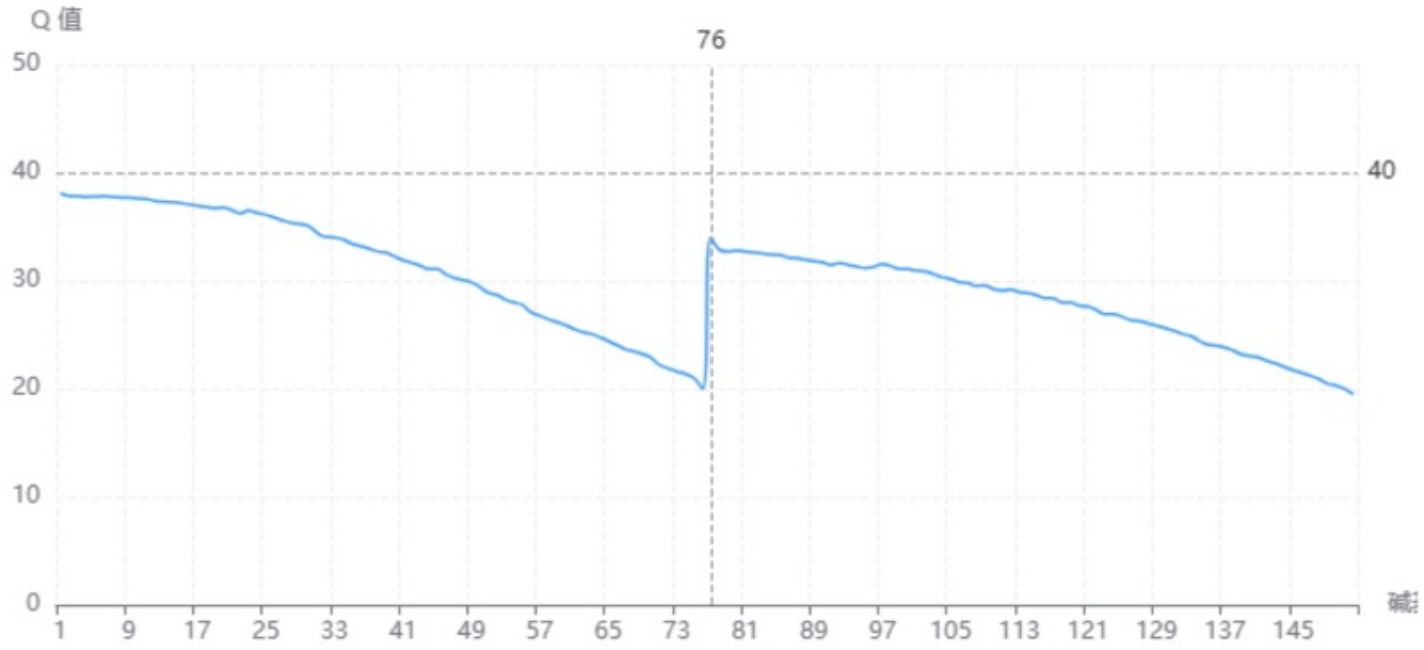
#	Sample	Raw Reads No	Raw Bases(bp)	Q30(bp)	GC(%)	N(%)	Q20(%)	Q30(%)
1	SRR307902	79304014	6027105064	4331995738	49.59	0.027712354701539474	80.24	71.88
2	SRR1039512	56272564	3545171532	3346065538	49.8	0.000012106603750793652	97.37	94.38
3	SRR1039513	33646176	2119709088	1958277231	48.75	0.000004944550324603171	96.45	92.38
4	SRR307901	80381908	6109025008	4438009536	47.18	0.02888854420946842	81.44	72.65
5	SRR1039509	42311414	2665619082	2537437023	49.67	0.0014545820728769844	97.44	95.19
6	SRR1039508	45871042	2889875646	2775797265	49.98	0.0014522572359880956	98.08	96.05

2.2数据过滤表

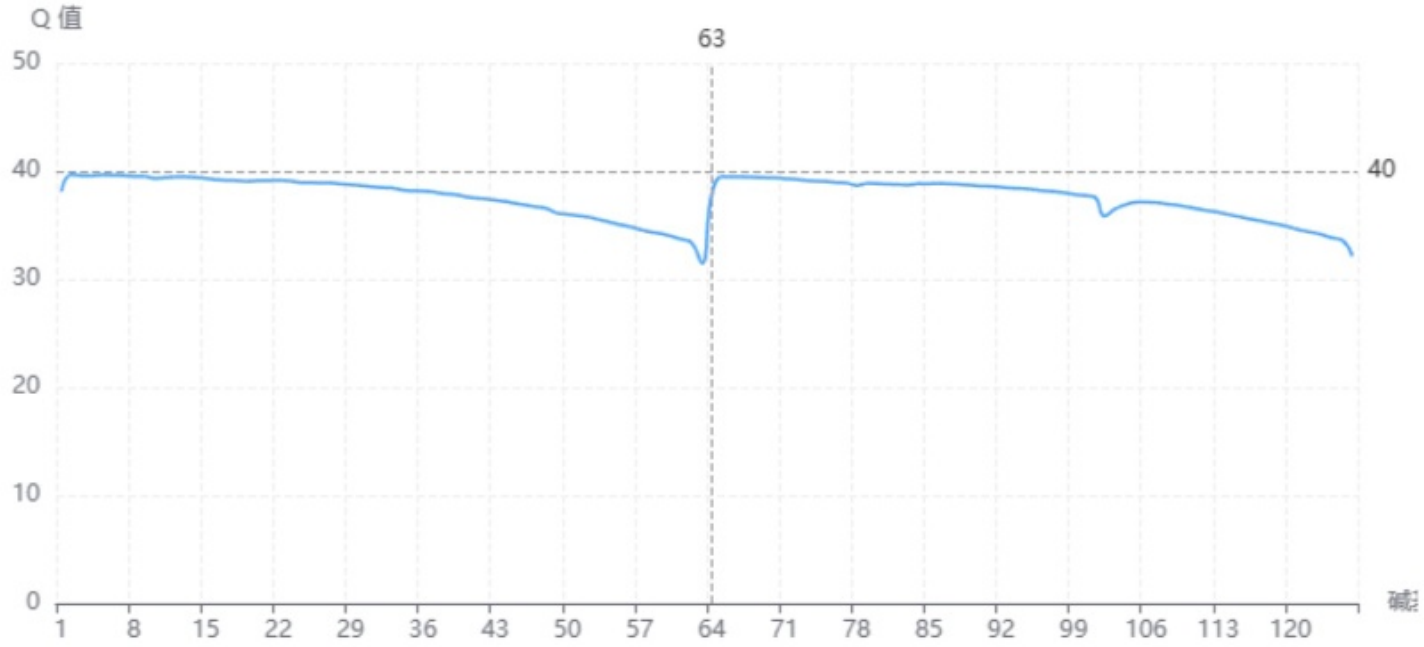
#	Sample	Clean Reads No	Clean Data(bp)	Clean Data(%)	Clean Reads(%)
1	SRR307902	60042206	4560405122	75.66	75.71
2	SRR1039512	54629390	3441115190	97.06	97.08
3	SRR1039513	32355496	2038090022	96.15	96.16
4	SRR307901	62875118	4777234152	78.2	78.22
5	SRR1039509	41183730	2594036776	97.31	97.33
6	SRR1039508	45052986	2837751394	98.2	98.22

2.3碱基质量分布

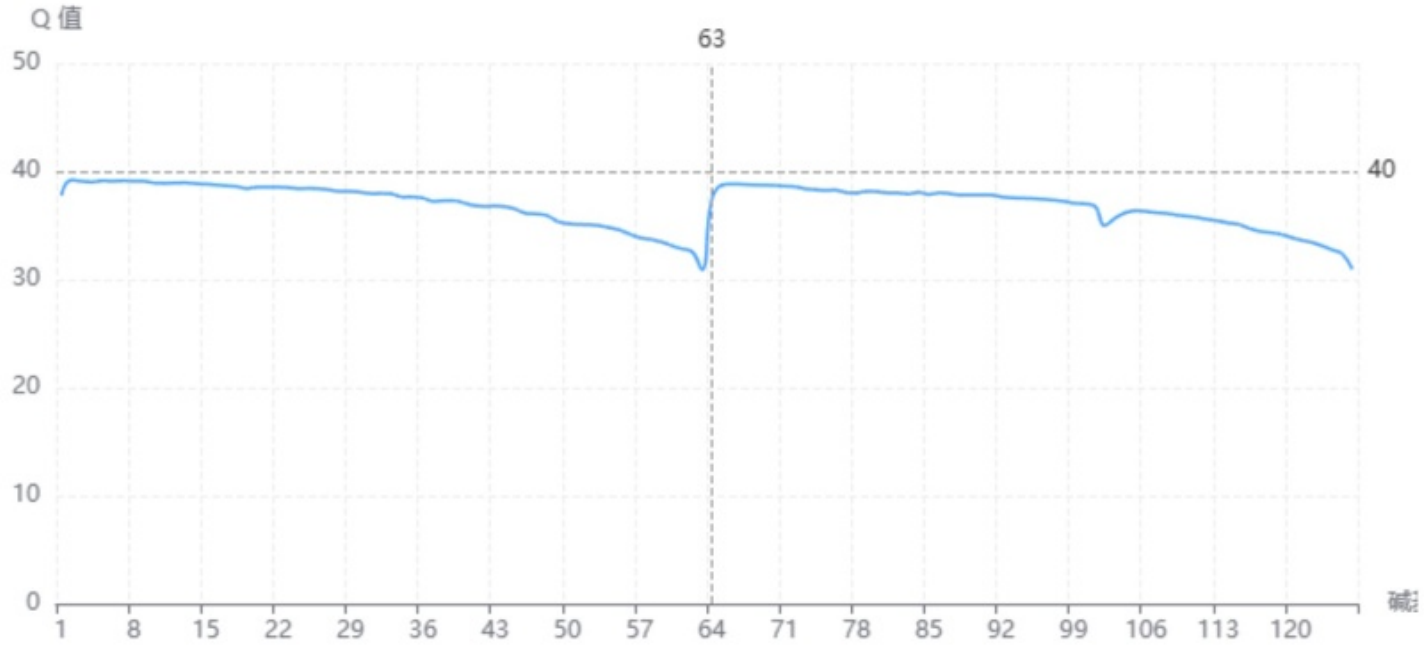
SRR307902



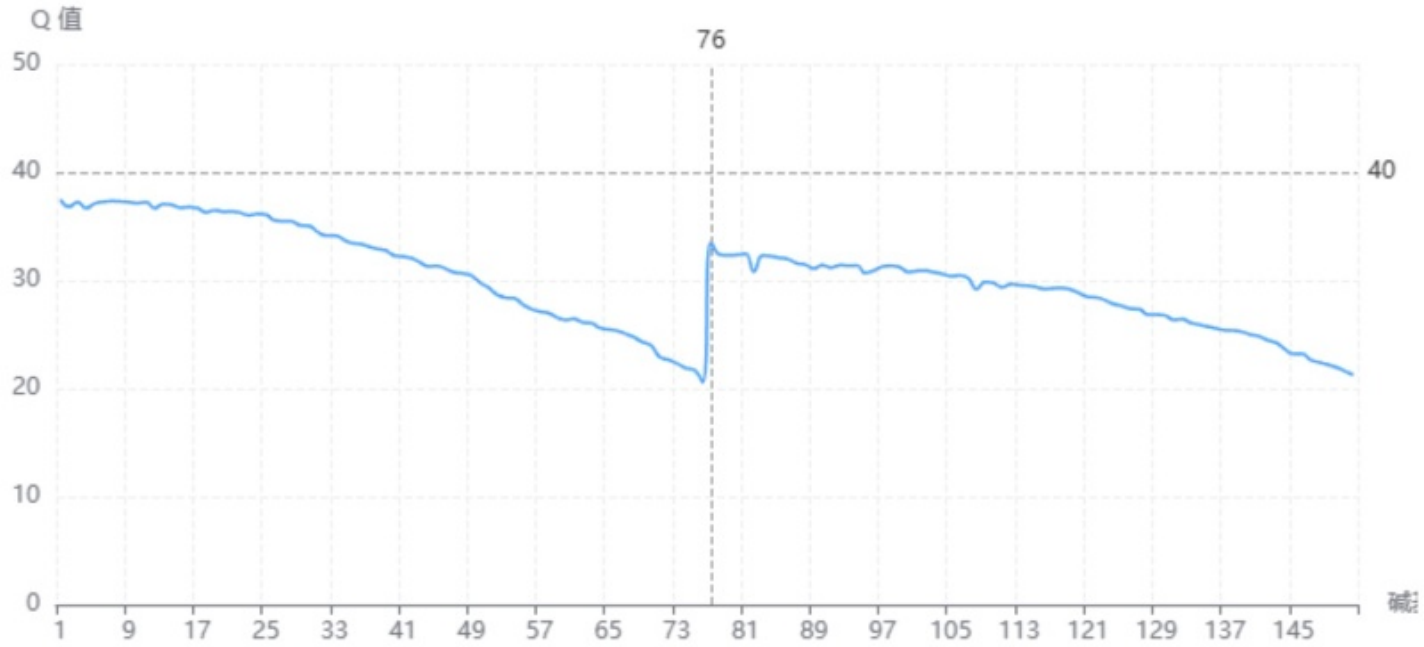
SRR1039512



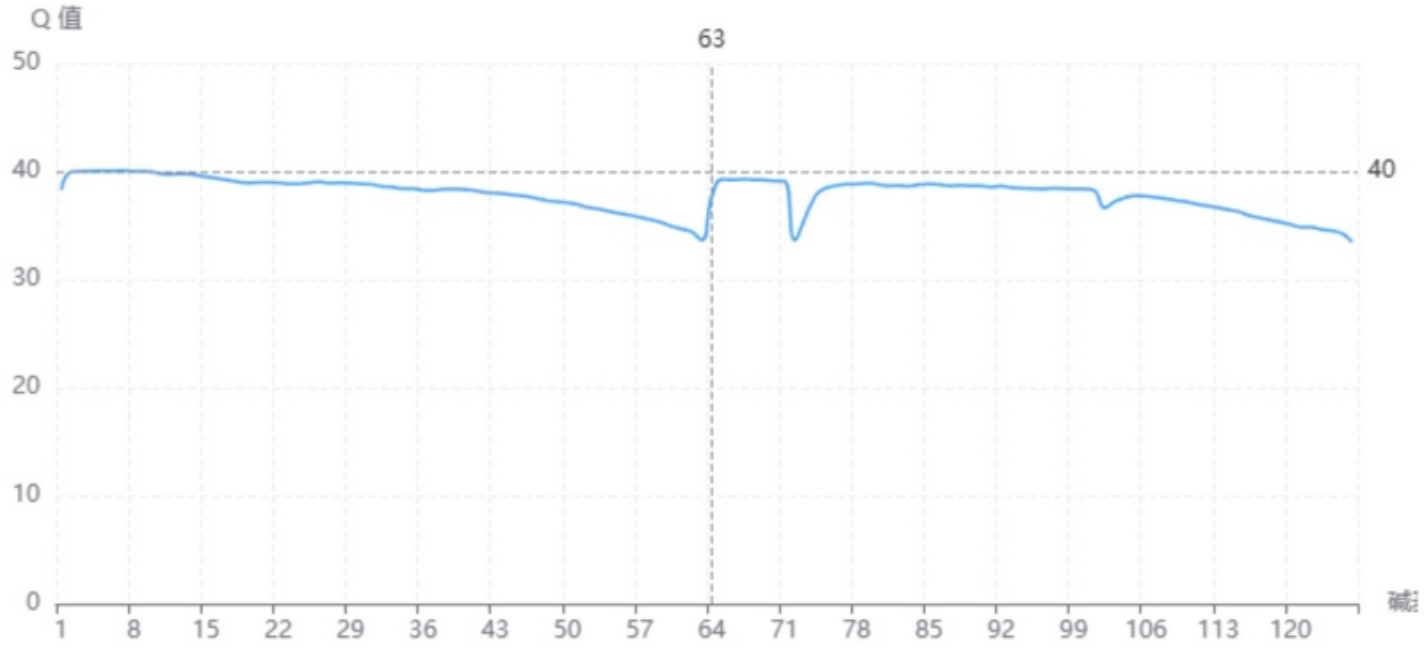
SRR1039513



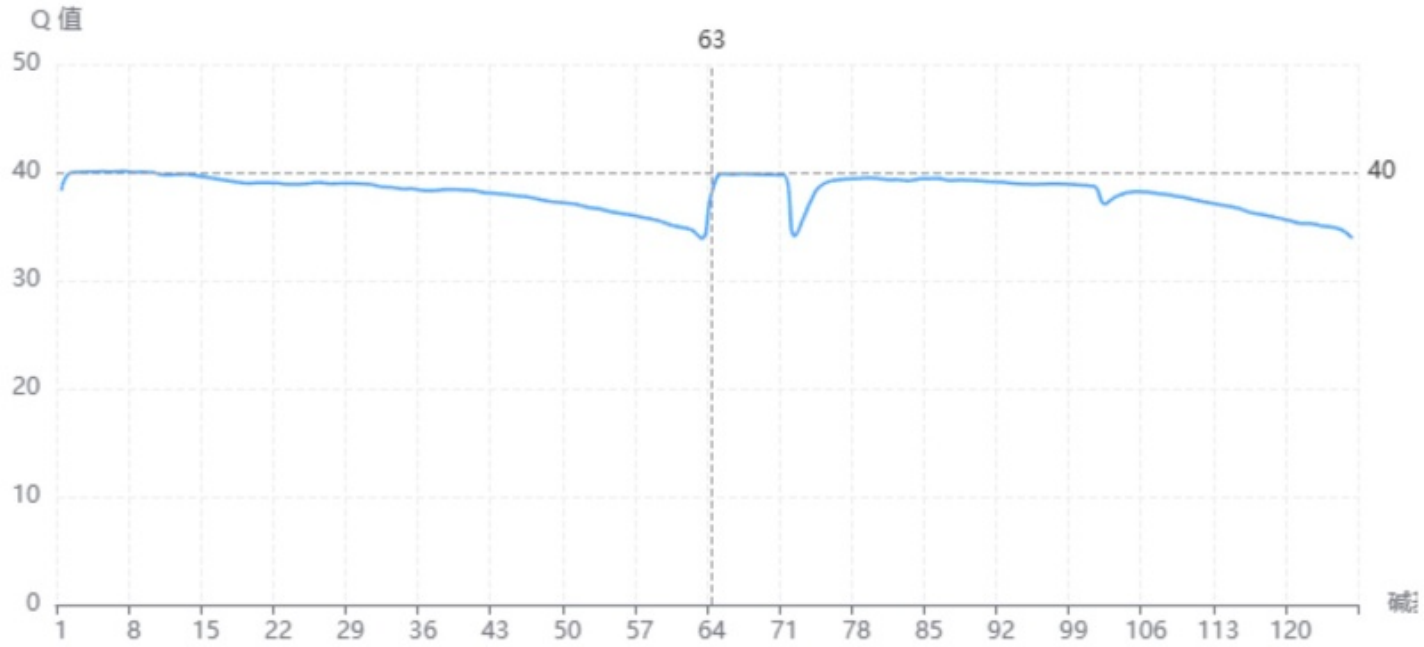
SRR307901



SRR1039509

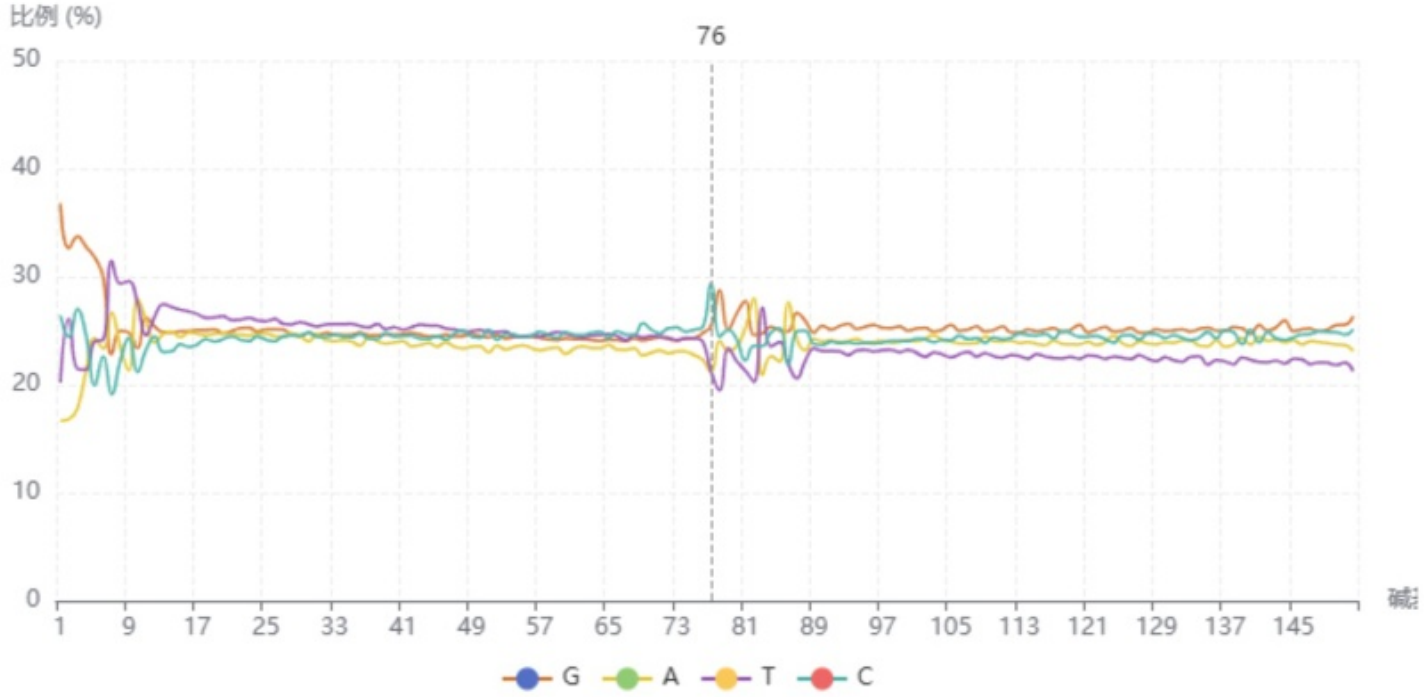


SRR1039508

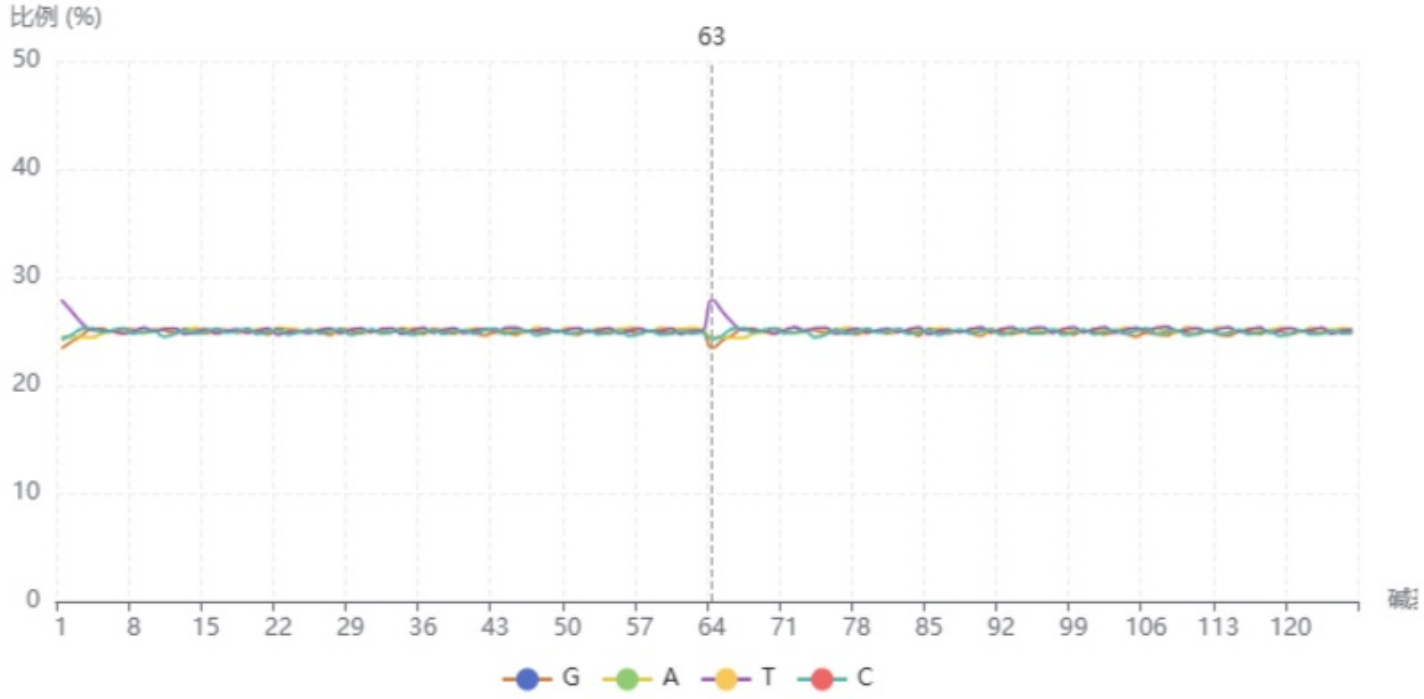


2.4碱基含量分布

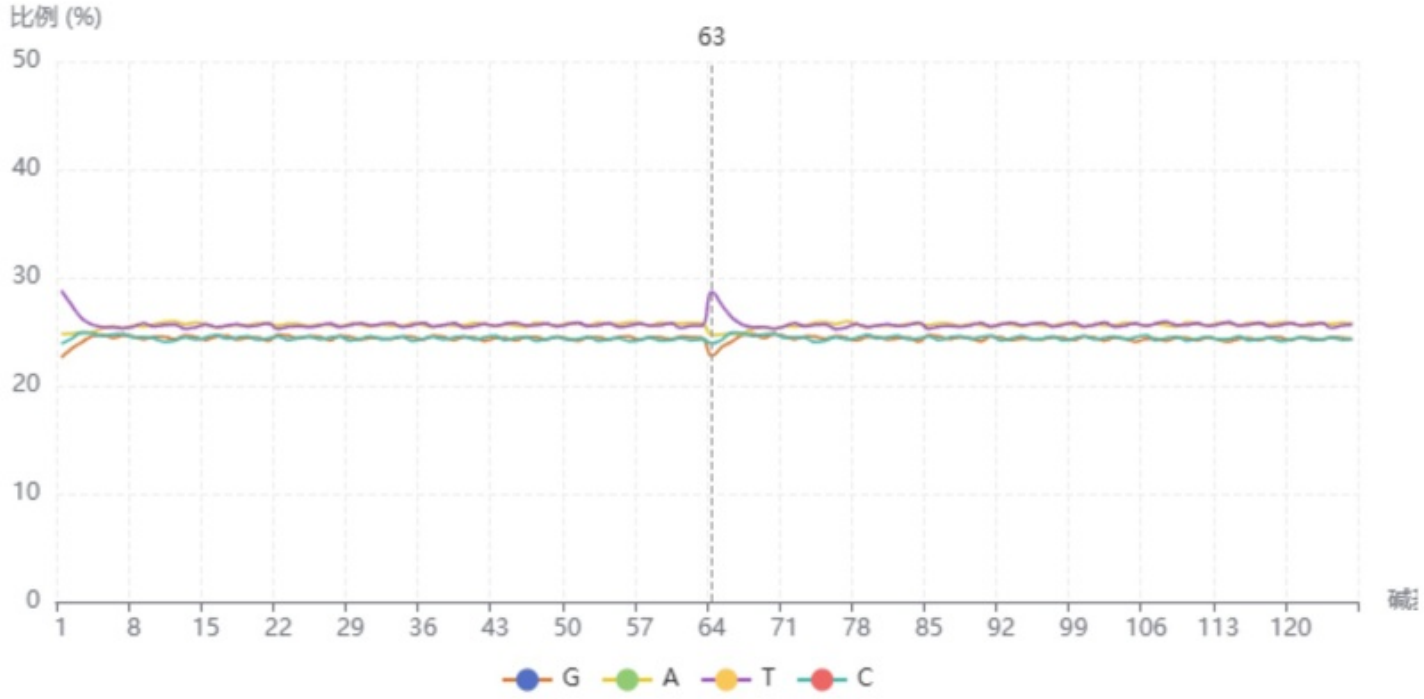
SRR307902



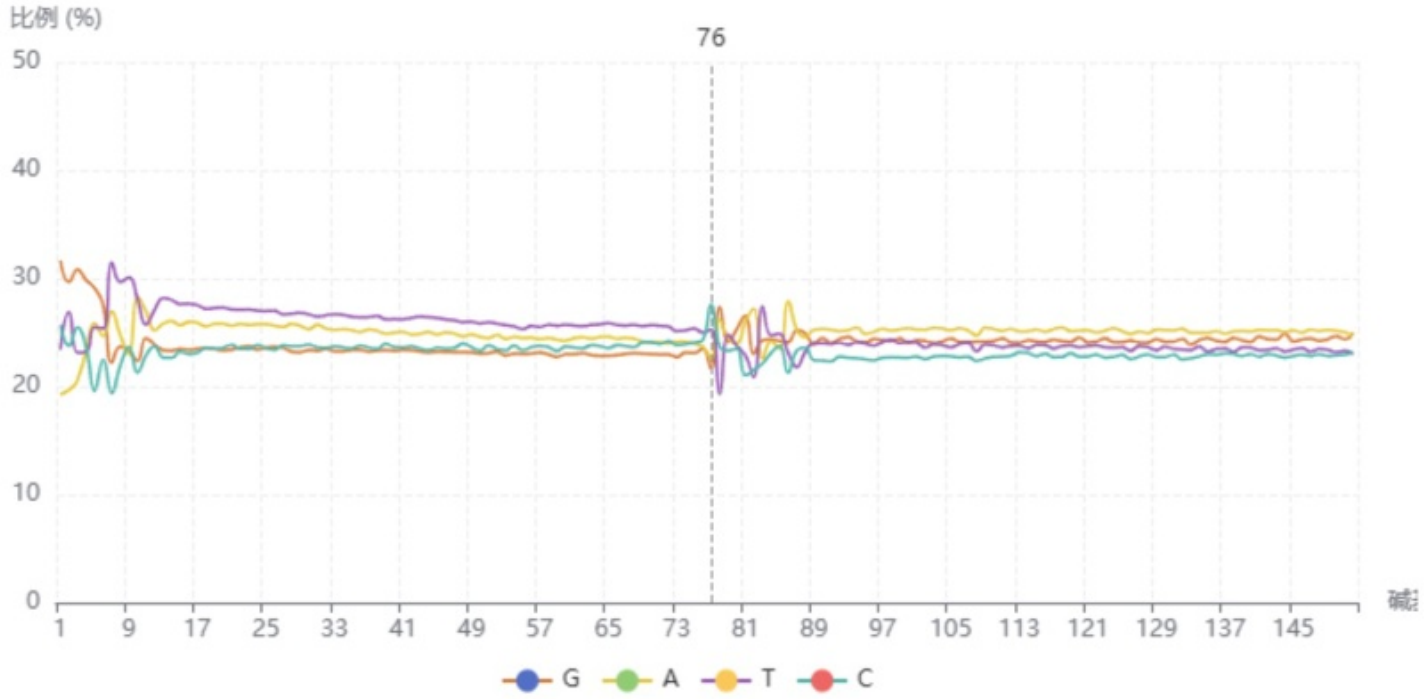
SRR1039512



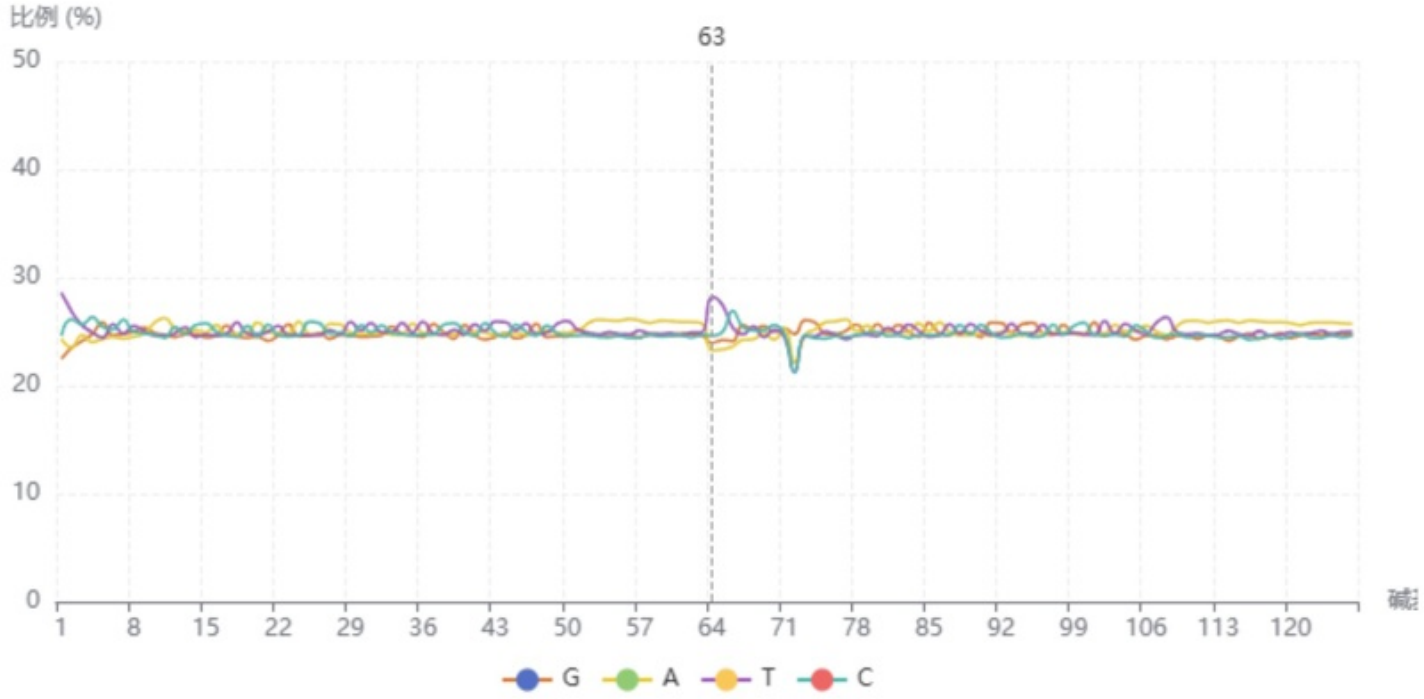
SRR1039513



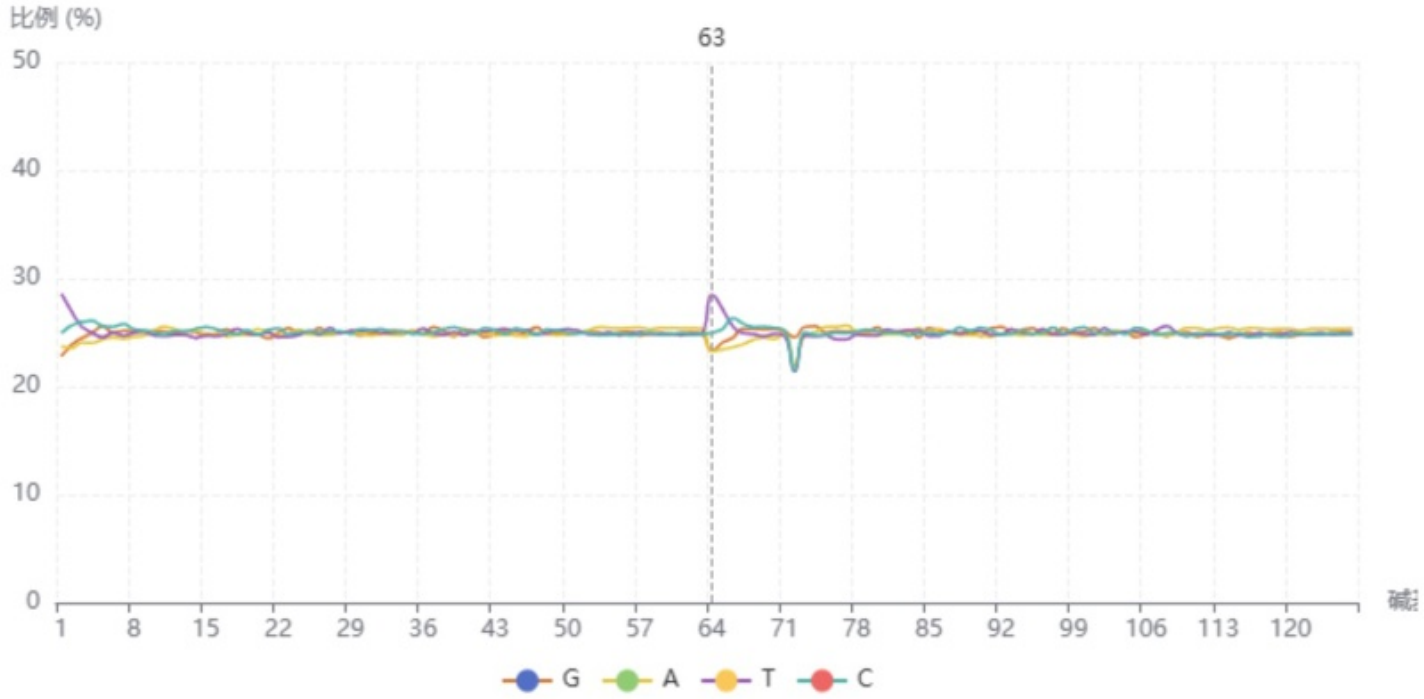
SRR307901



SRR1039509

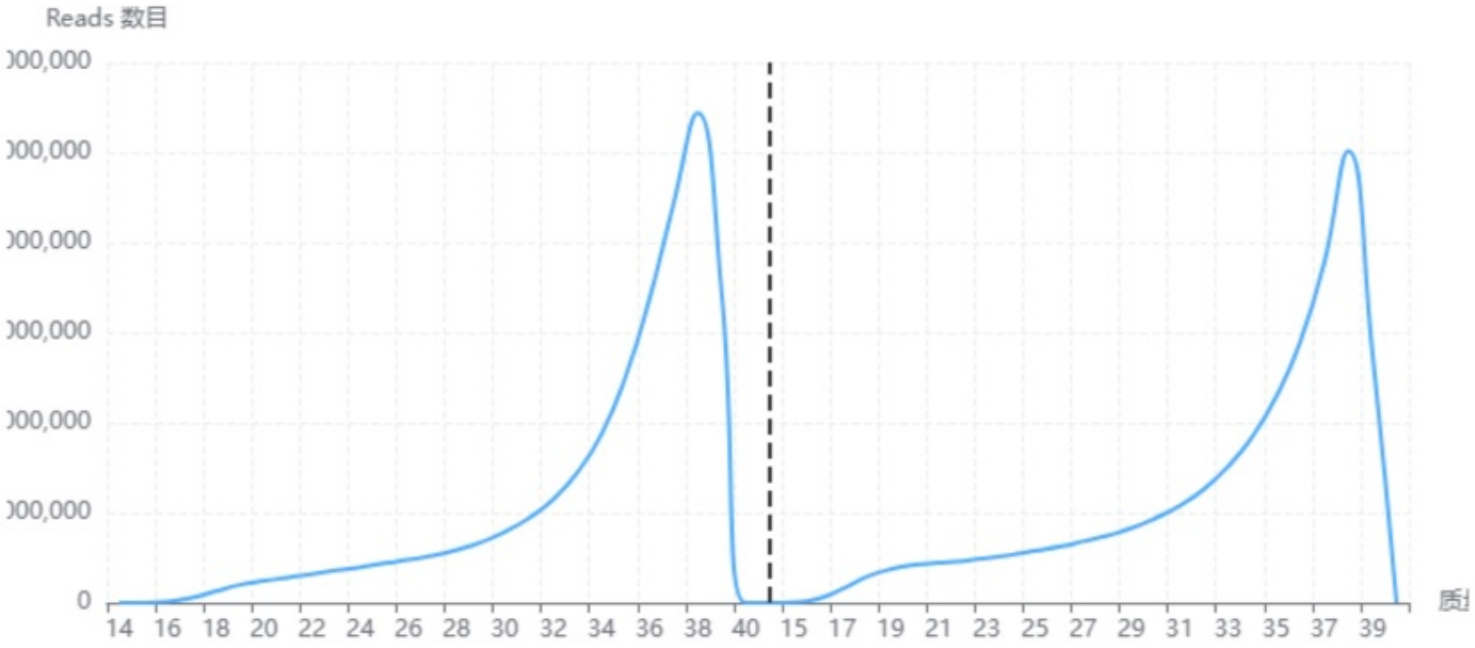


SRR1039508

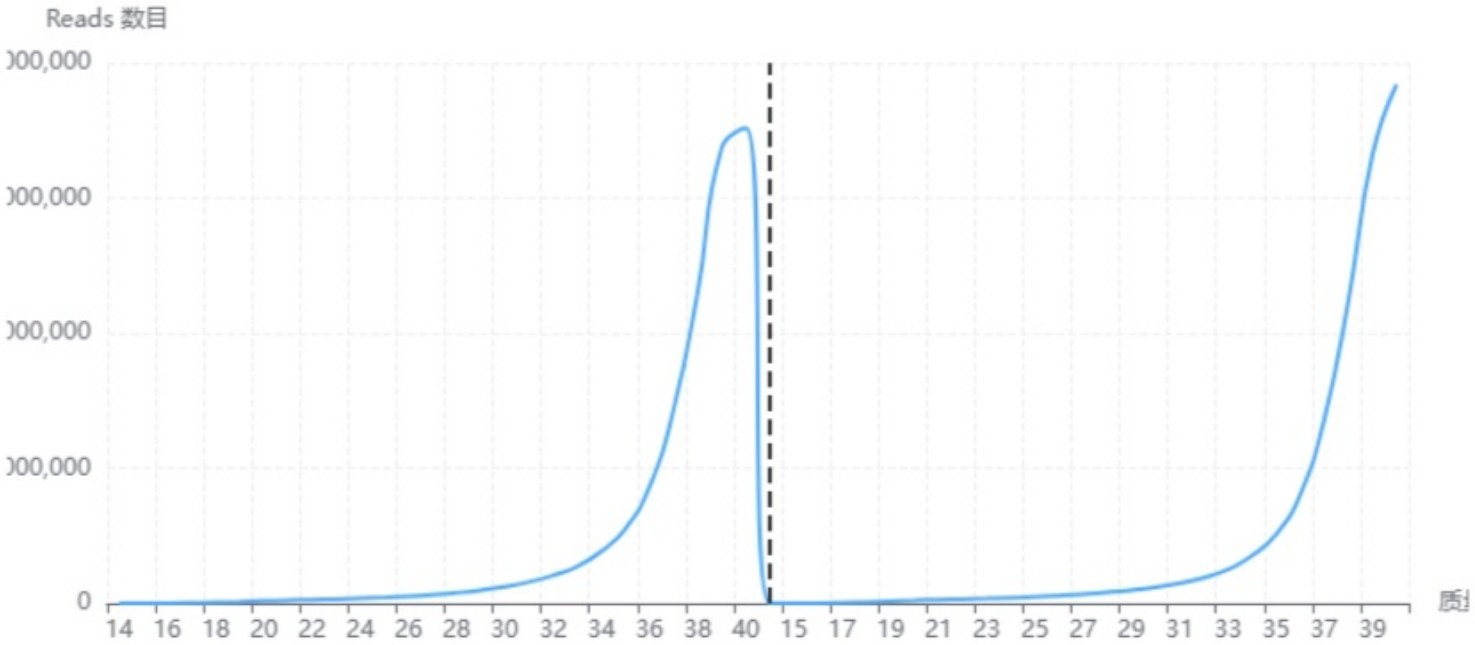


2.5 Reads平均质量分布

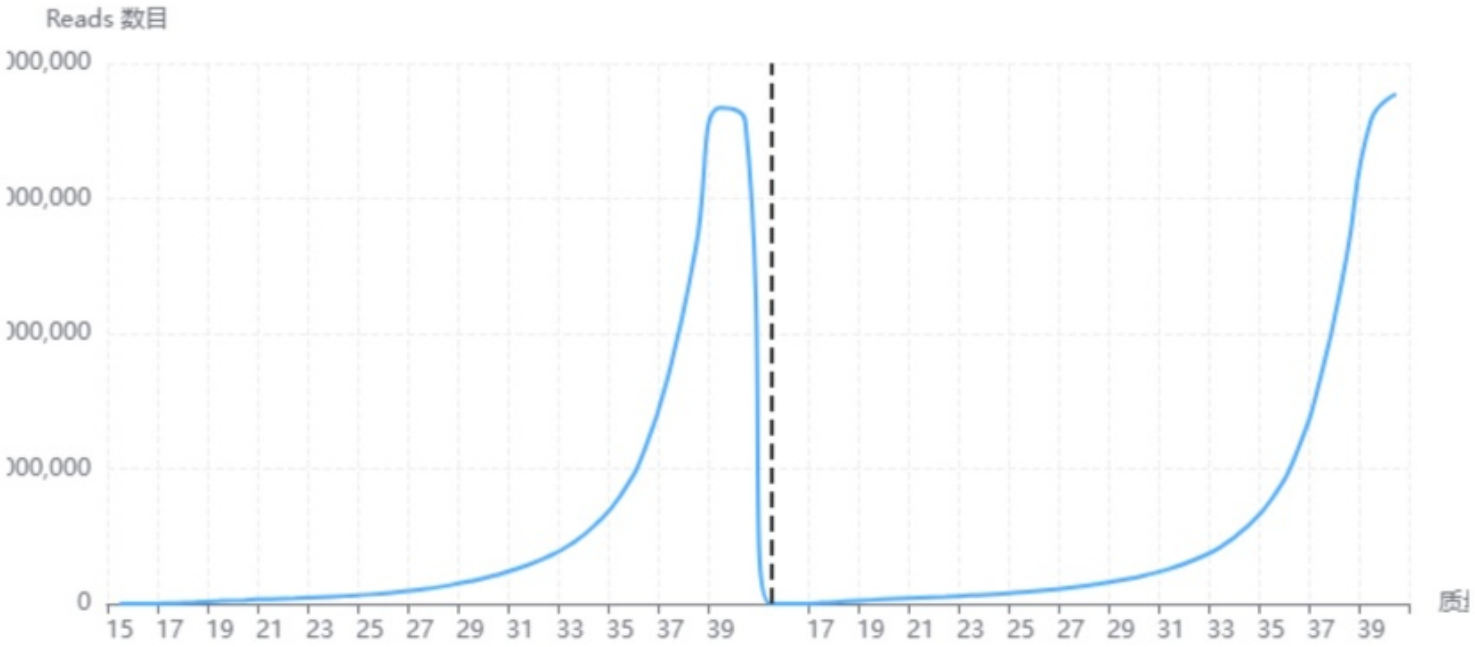
SRR307902



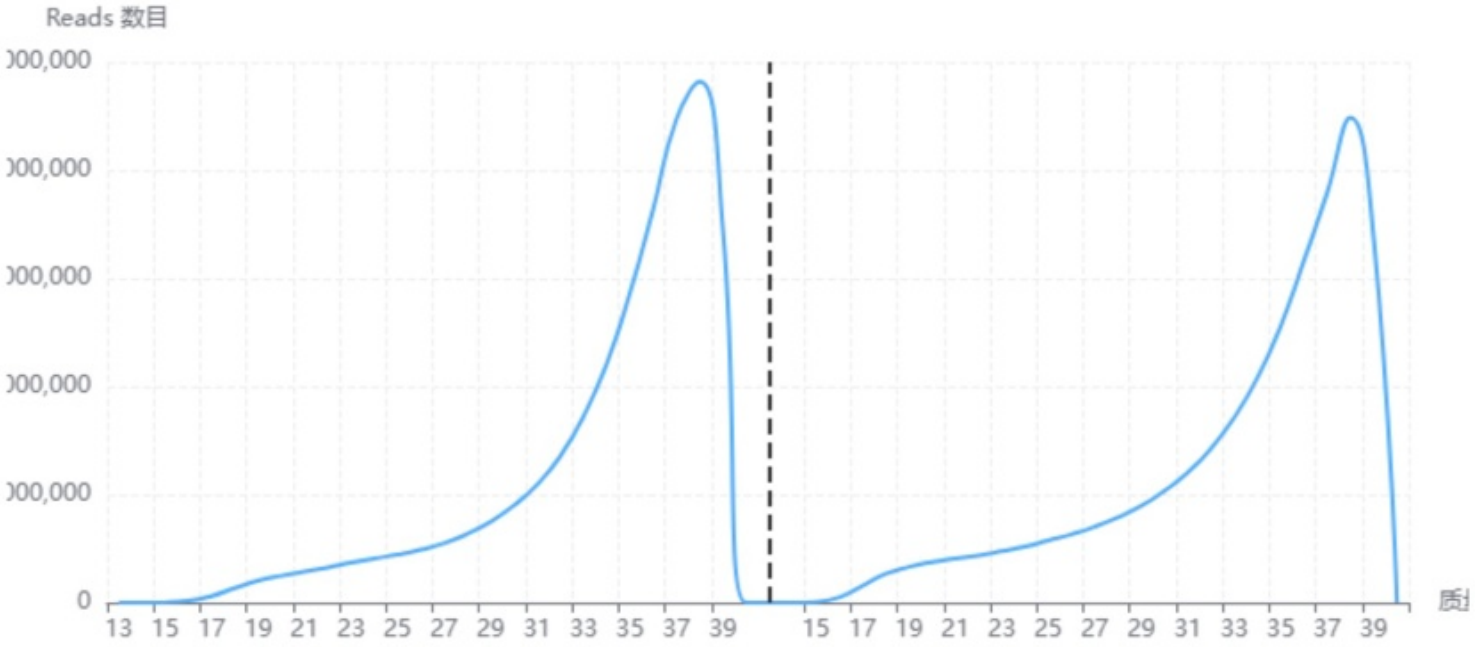
SRR1039512



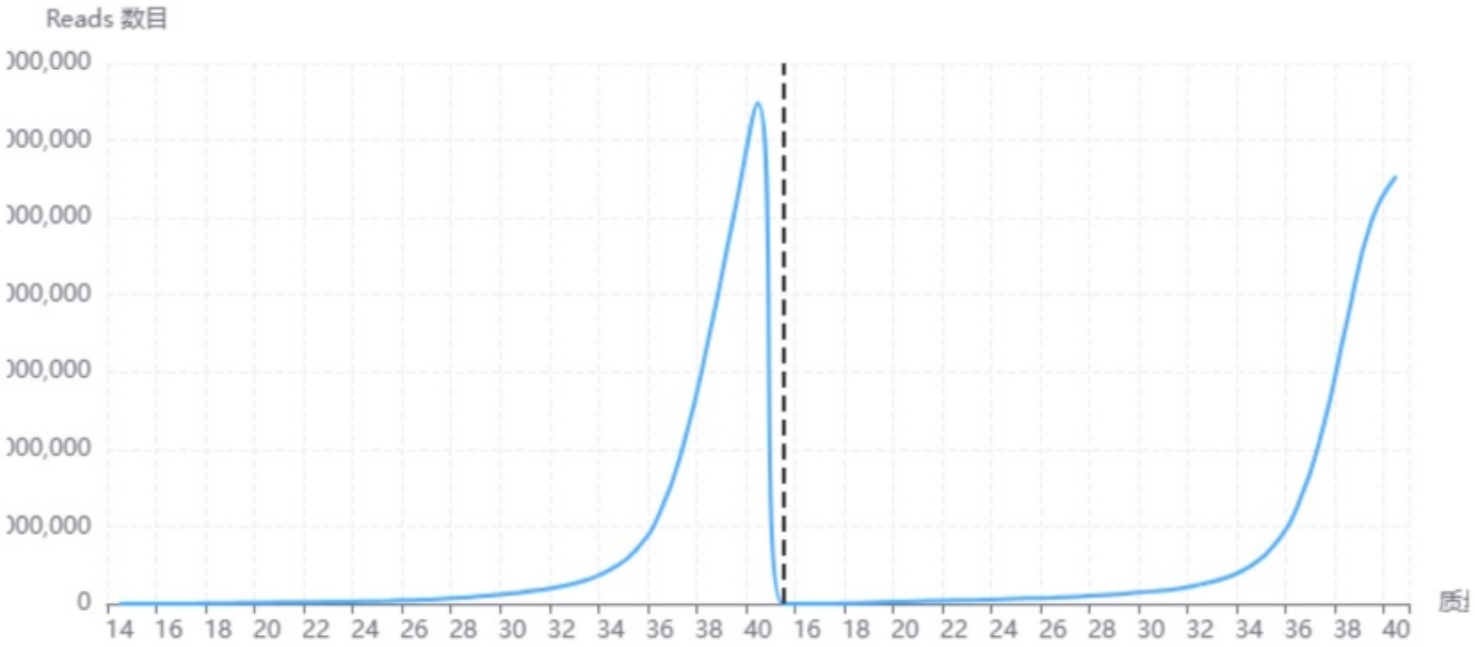
SRR1039513



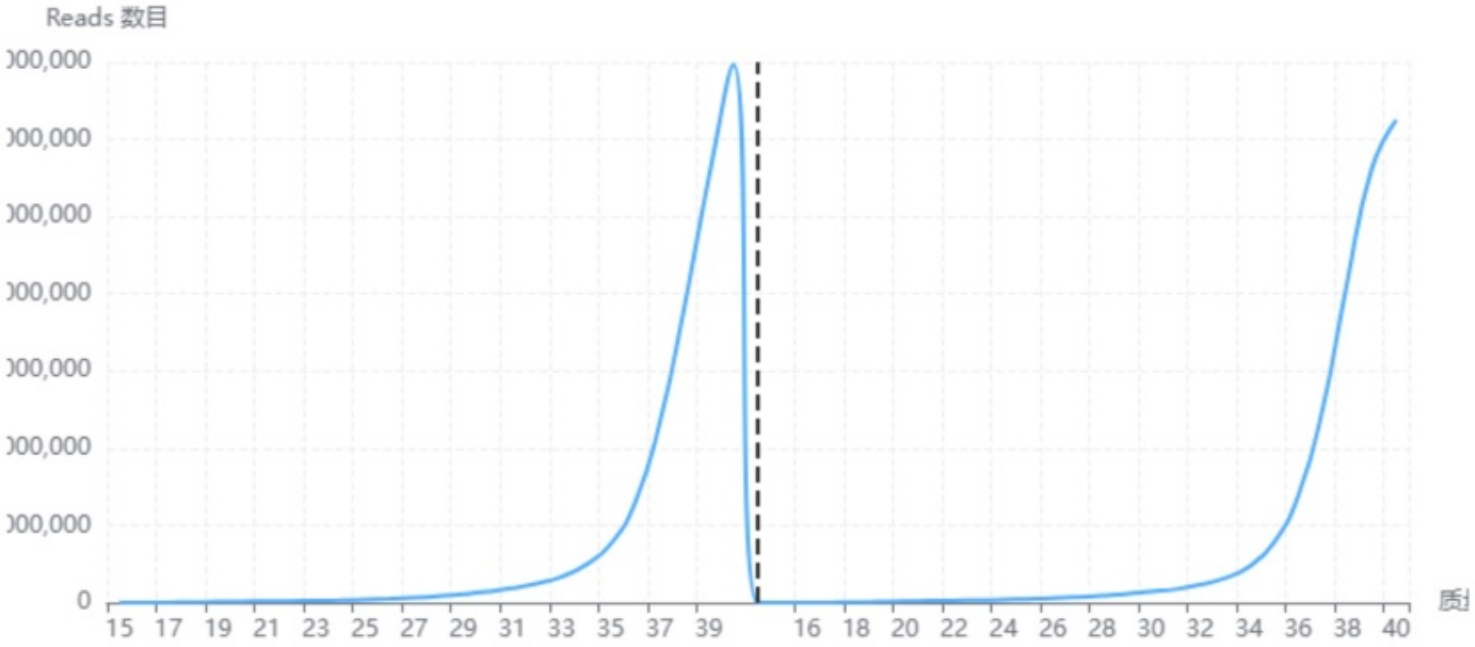
SRR307901



SRR1039509



SRR1039508



三、序列比对及统计

将测序数据与参考基因组进行比对，统计比对结果。

#	Sample	Clean Reads	Total Mapped	Multiple Mapped	Uniquely Mapped
1	SRR307901	62875118	59003736	2735127	56268609
2	SRR1039513	32355496	31851706	1054016	30797690
3	SRR1039508	45052986	44055154	1755630	42299524
4	SRR1039512	54629390	53876690	1904986	51971704
5	SRR307902	60042206	56920184	3113926	53806258
6	SRR1039509	41183730	40053061	1507007	38546054

四、基因表达定量

对基因表达水平进行定量分析，统计表达量分布。

4.1表达量矩阵

#	Gene ID	Gene Length	Chromosome	SRR307901 Read Count	SRR307901 FPKM	SRR307902 Read Count	SRR30
1	7SK	6132	5	63	0.27286645618698724	87	0.3796
2	A1BG	4006	19	35	0.23518296013879683	21	0.1521
3	A1BG-AS1	2793	19	114	0.8696013936592092	170	1.1963
4	A2M	6384	12	30	0.13123650434437928	61	0.2663
5	A2M-AS1	2187	12	1	0.013306858441742548	2	0.0277
6	A4GALT	2943	22	85	0.6647613400955497	112	0.8540
7	A4GNT	1771	3	6	0.0957994147600677	6	0.1001
8	AAAS	4138	12	1481	3.0449221365422314	1614	3.2135
9	AACS	16089	12	1471	1.5123988364115908	1688	1.6898
10	AADACL4	1575	1	0	0	4	0.0757

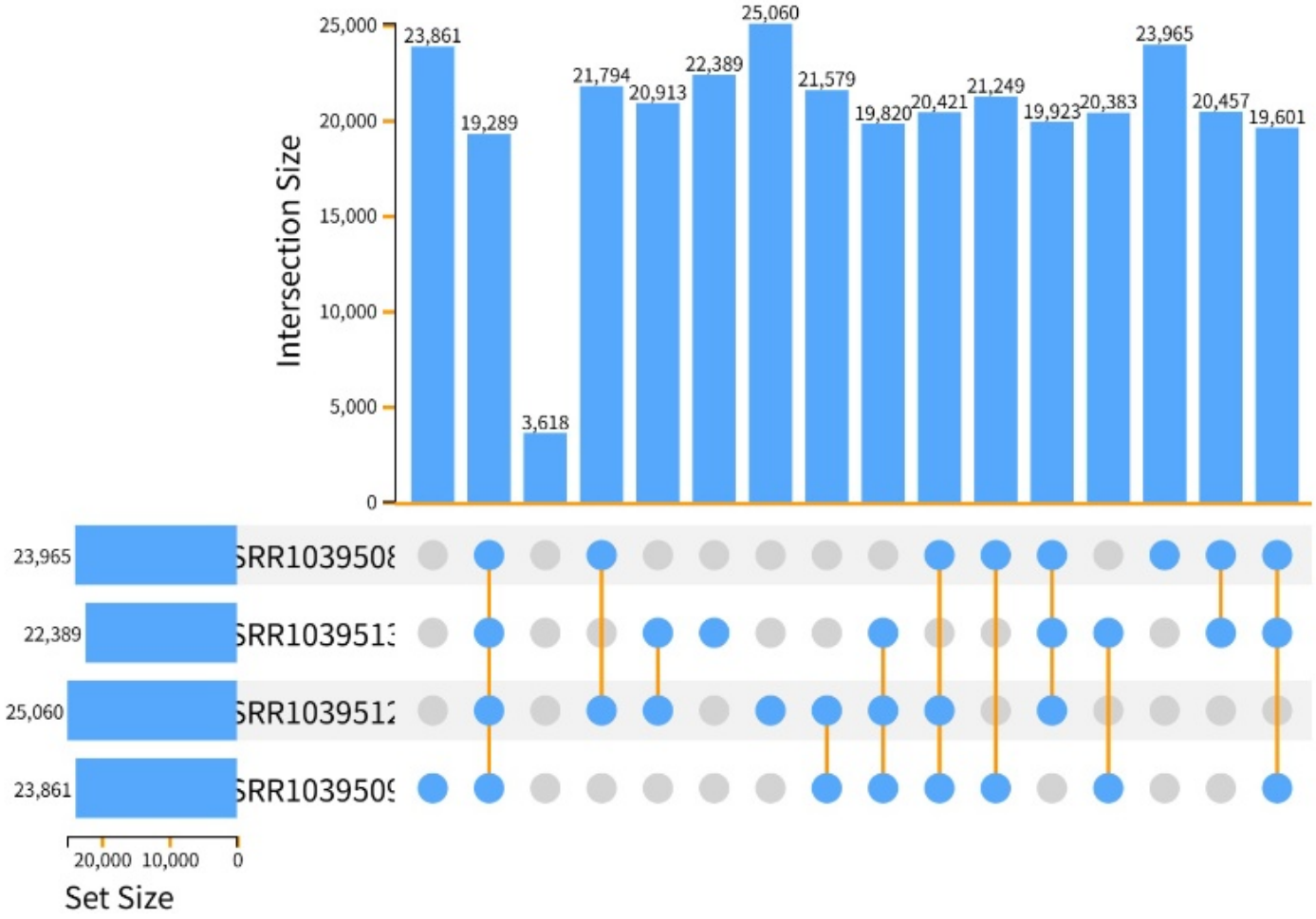
4.2表达量区间统计

#	Type	SRR307901	SRR307902	SRR1039508	SRR1039509	SRR1039512	SRR1039513
1	0	346	392	321	341	241	495
2	(0,0.01]	153	143	62	62	56	60
3	(0.01,0.1]	1498	1447	1165	1343	1214	1332
4	(0.1,1]	4641	4719	4818	4902	4948	4921
5	(1,10]	7728	7779	8089	7805	8285	7907
6	(10,100]	3539	3418	3467	3446	3206	3219
7	(100,1000]	315	319	293	316	262	278
8	>1000	20	23	25	25	28	28

4.3表达量区间统计图



4.4基因表达丰度图（UpSet图）



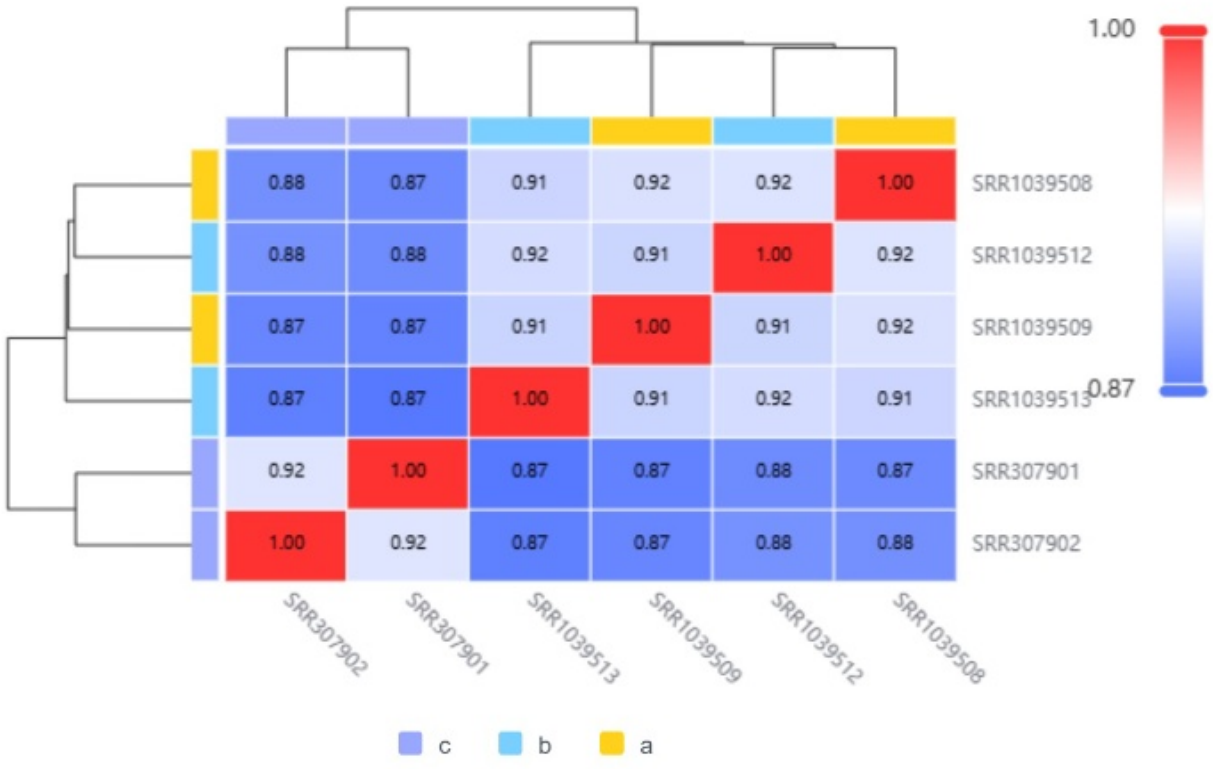
五、样本间关系分析

通过相关性分析和主成分分析（PCA）评估样本间关系。

5.1相关性分析

#	Sample	SRR307902	SRR1039513	SRR1039512	SRR307901	SRR1039508	SRR1(
1	SRR307902	1	0.86891127469412	0.878084470703919	0.919961047716391	0.877260768532357	0.8722
2	SRR1039513	0.86891127469412	1	0.915481906703633	0.865533406953067	0.911864230221875	0.9117
3	SRR1039512	0.878084470703919	0.915481906703633	1	0.875126808000453	0.919243297548793	0.9121
4	SRR307901	0.919961047716391	0.865533406953067	0.875126808000453	1	0.874390388987008	0.8699
5	SRR1039508	0.877260768532357	0.911864230221875	0.919243297548793	0.874390388987008	1	0.9176
6	SRR1039509	0.872279784534309	0.911722757459122	0.912107295890324	0.869910125536506	0.917681625855715	1

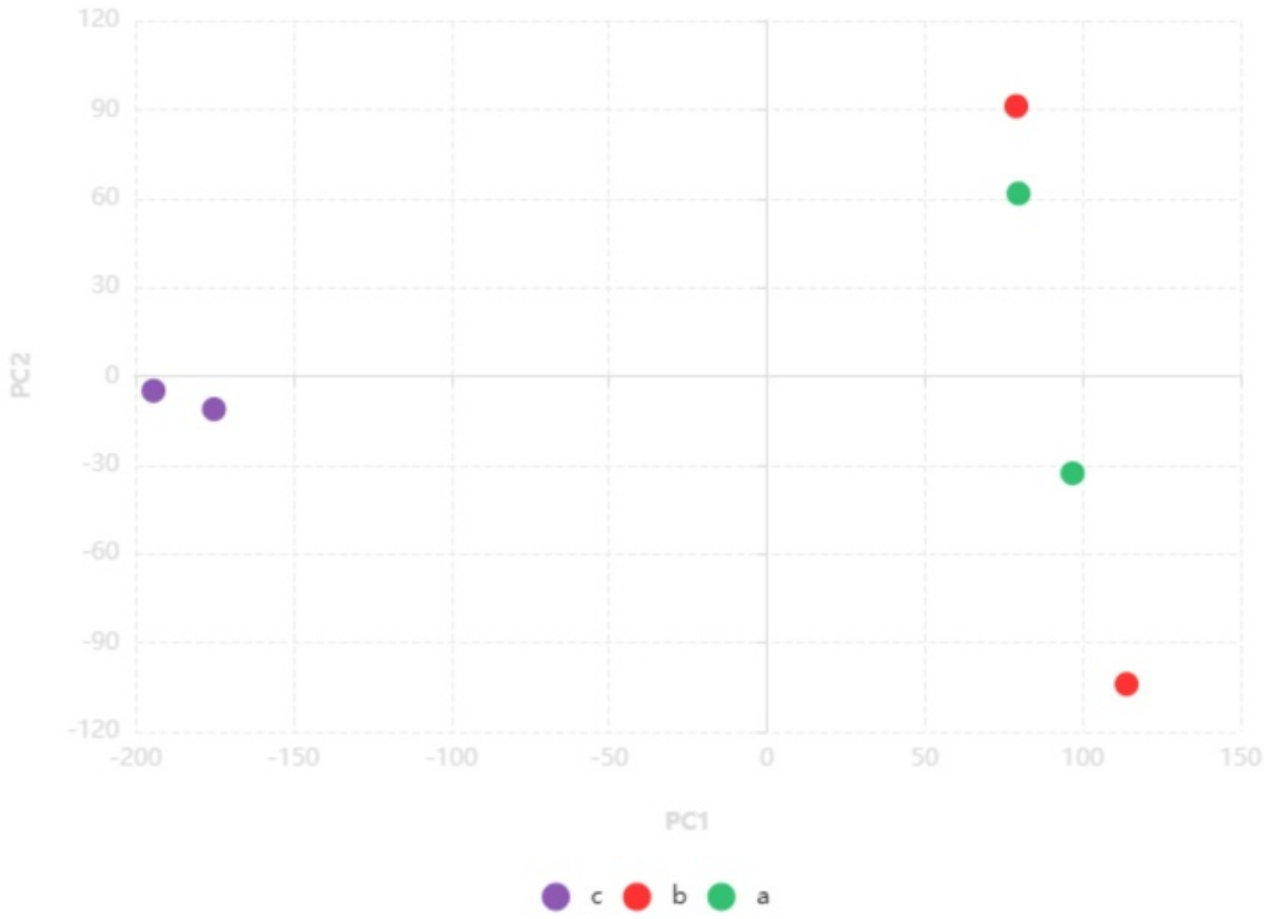
5.2相关性分析热图



5.3PCA分析

#	ID	Group	PC1	PC2	PC3
1	SRR307902	c	-175.12967783324	-11.1290511958736	-1.30643475231391
2	SRR1039513	b	113.926555836651	-103.975208834429	56.0349364789263
3	SRR1039512	b	78.9830021639837	91.1416890200405	75.630954914005
4	SRR307901	c	-194.298245897876	-4.92243857346074	3.27084429361393
5	SRR1039508	a	79.7054834997033	61.6689985853992	-57.6610868011166
6	SRR1039509	a	96.8128822307775	-32.7839890016766	-75.9692141331147

5.4PCA分析图



六、差异表达分析

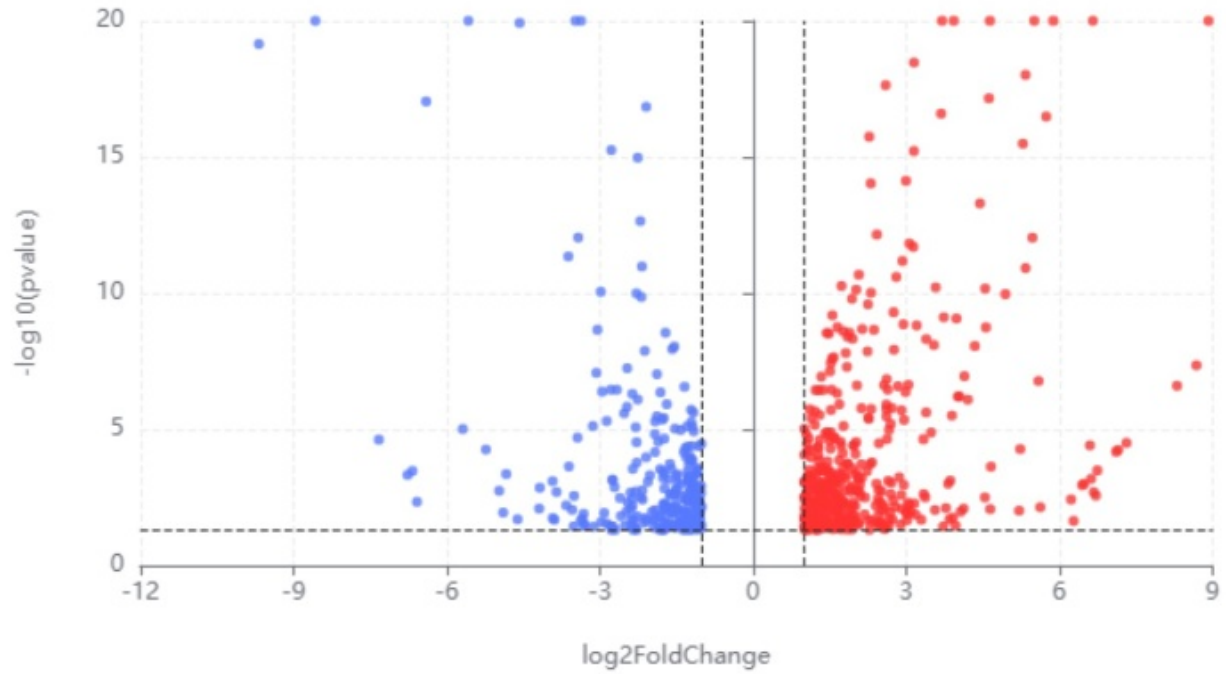
鉴定不同分组间的差异表达基因，并进行统计分析。

6.1差异表达统计

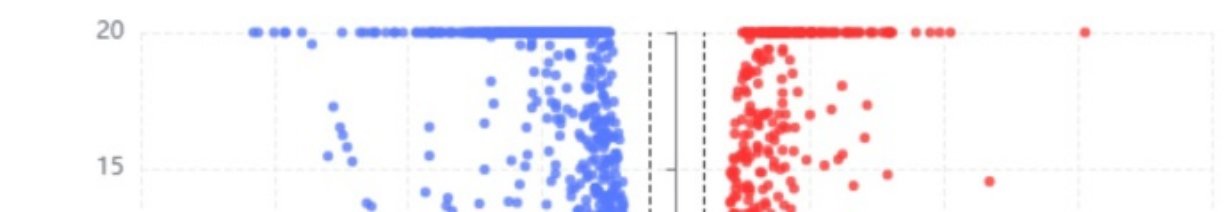
#	ID	baseMean	foldChange(b_vs_a)	log2FoldChange	pval	padj
1	A2M-AS1	56.6852770210279	0.4645372112890166	-1.10613392800624	0.0353100825807863	0.54936944718
2	ABC7-42389800N19.1	507.664403519002	0.4299768322977487 5	-1.21766916723447	0.000126304782083485	0.01029383973
3	ABCA13	50.9893795892457	2.6941333355278743	1.42982125308108	0.016375454847832	0.35478796024
4	ABCA3	266.831411200342	2.048539581103184	1.03459576829276	0.00288544886317959	0.11158561778
5	AC005682.5	84.7166214286712	3.2260370527176057	1.68976300869348	0.000136659466413488	0.01099031587
6	AC005789.11	31.8537481313671	2.540753636045178	1.34525649137421	0.0390597933380054	0.58076884536
7	AC009302.2	19.2987474161531	3.299831912575207	1.72239253808615	0.0373337322806147	0.56473341258
8	AC011043.1	246.236974949008	0.1463793243565642	-2.77221630325145	5.58305607374346e-16	4.05725337425
9	AC012309.5	9.98248333176237	9.839500163671625	3.29858503002511	0.020858978916152	0.41240224113
10	AC013449.1	62.4822524738826	0.42554785099766573	-1.23260672909697	0.0127051310115426	0.29879099855

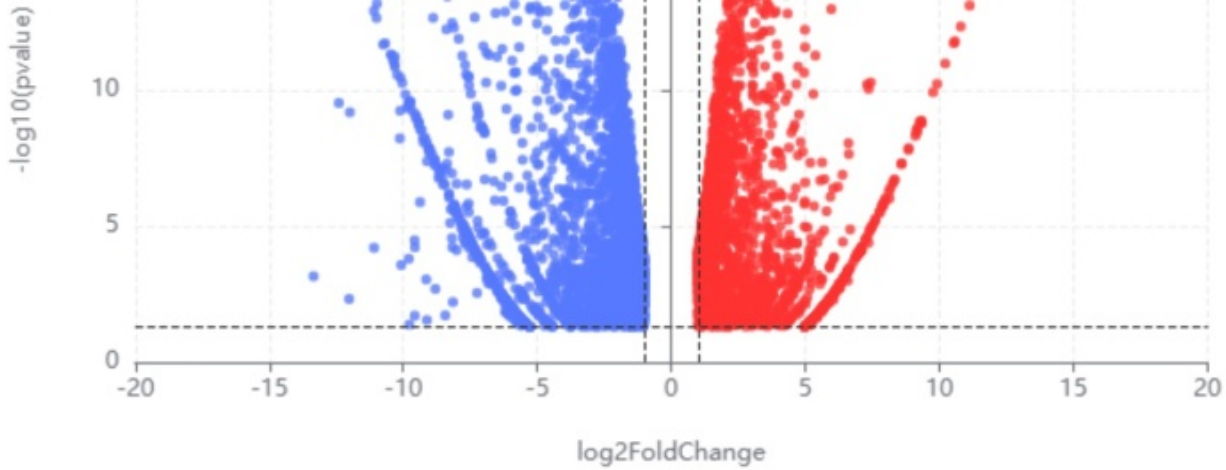
6.2差异表达火山图

b_vs_a

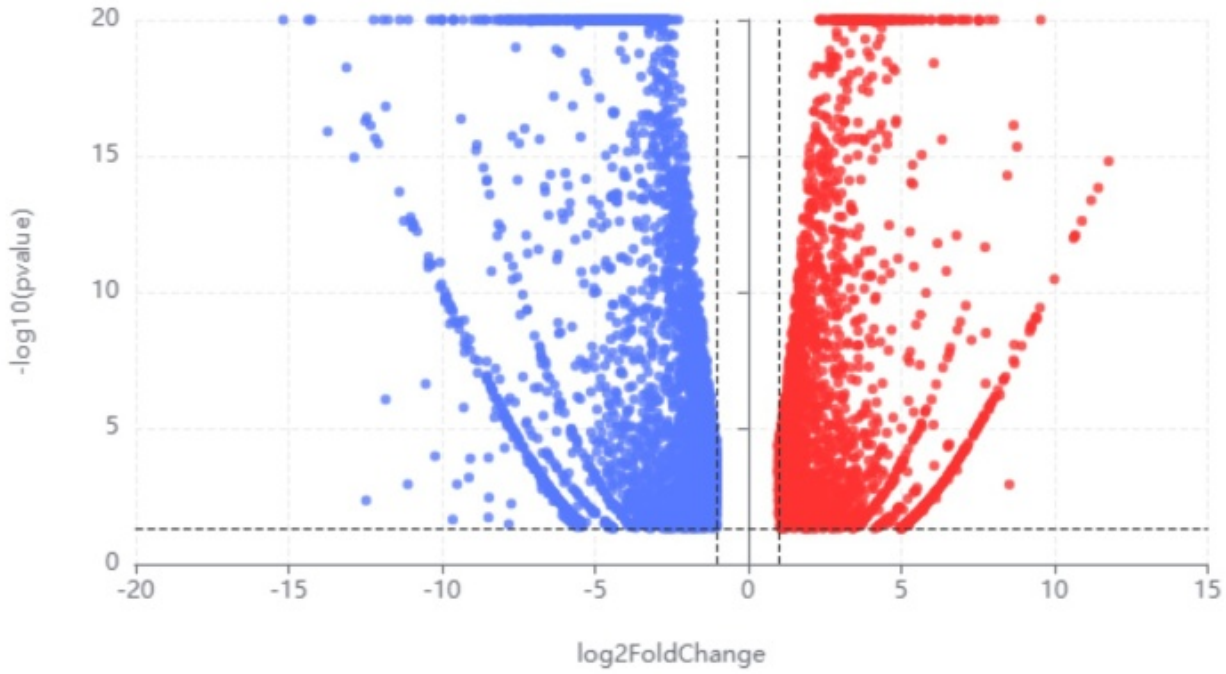


c_vs_b




-log10(pvalue)
10
5
0
-20
-15
-10
-5
0
5
10
15
20
log2FoldChange

c_vs_a


-log10(pvalue)
20
15
10
5
0
-20
-15
-10
-5
0
5
10
15
log2FoldChange

七、功能富集分析

对差异表达基因进行GO和KEGG通路富集分析。

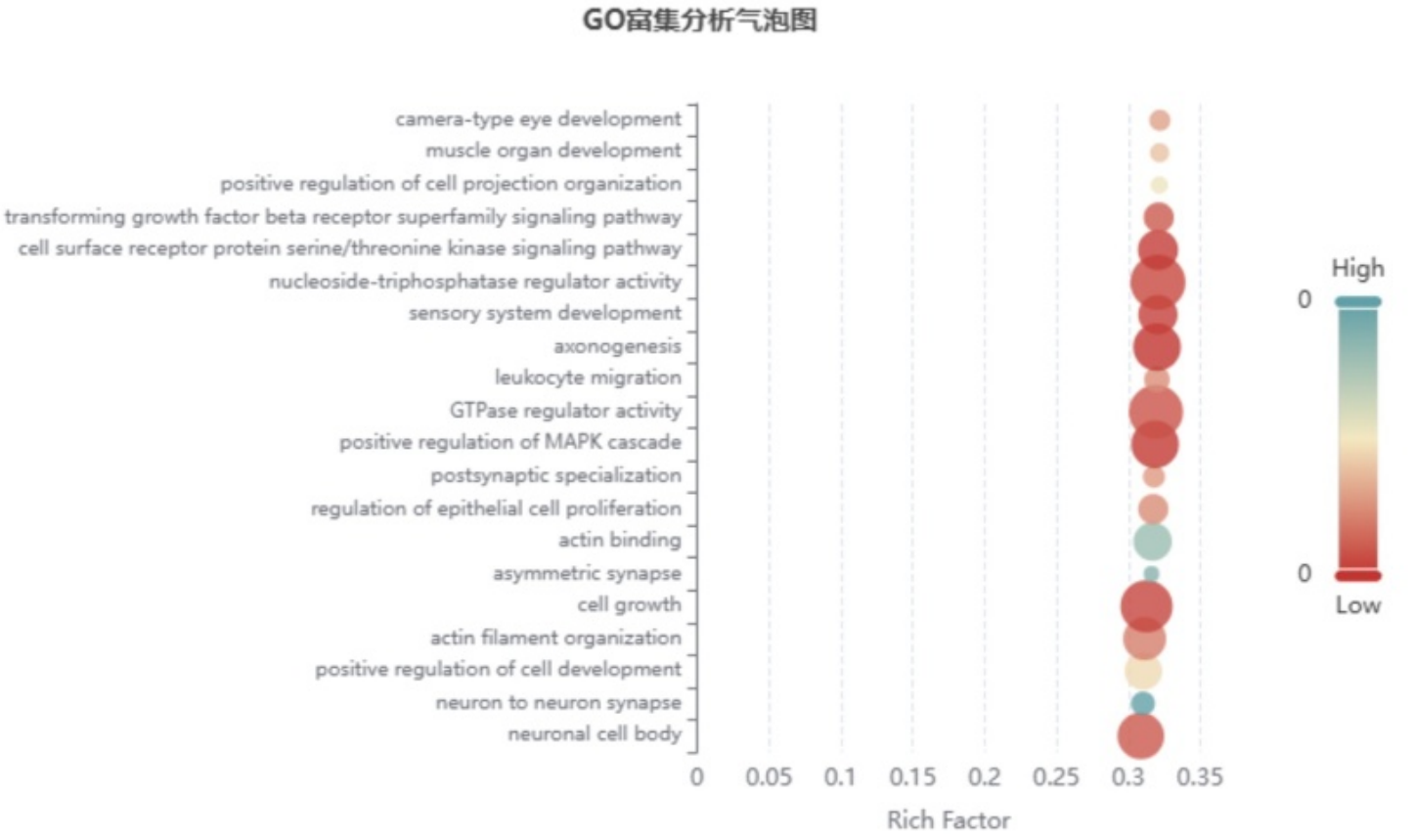
7.1GO富集分析

#	GO_ID	Category	Term	Up	Down	DEG	Total	Pvalue	adjustPva
1	GO:0031012	CC	extracellular matrix	50	139	189	19944	3.63341854280768e-19	1.7636651
2	GO:0030312	CC	external encapsulating structure	50	139	189	19944	4.67815692103341e-19	1.7636651
3	GO:0048732	BP	gland development	58	105	163	18860	1.59092791429772e-13	1.0053073
4	GO:0036293	BP	response to decreased oxygen levels	44	81	125	18860	1.6851228552784e-12	4.0318628
5	GO:0072001	BP	renal system development	47	79	126	18860	1.91416181747646e-12	4.0318628
6	GO:0060541	BP	respiratory system development	31	64	95	18860	3.60156447800792e-12	5.6895714
7	GO:0005201	MF	extracellular matrix structural constituer	23	56	79	18662	4.55595916276062e-12	5.7177287
8	GO:0070482	BP	response to oxygen levels	47	84	131	18860	6.86375435846349e-12	8.6744127
9	GO:0031589	BP	cell-substrate adhesion	42	92	134	18860	9.27740868147953e-12	9.0214245
10	GO:0060537	BP	muscle tissue development	66	91	157	18860	1.1158235114756e-11	9.0214245

7.2GO富集分析柱状图



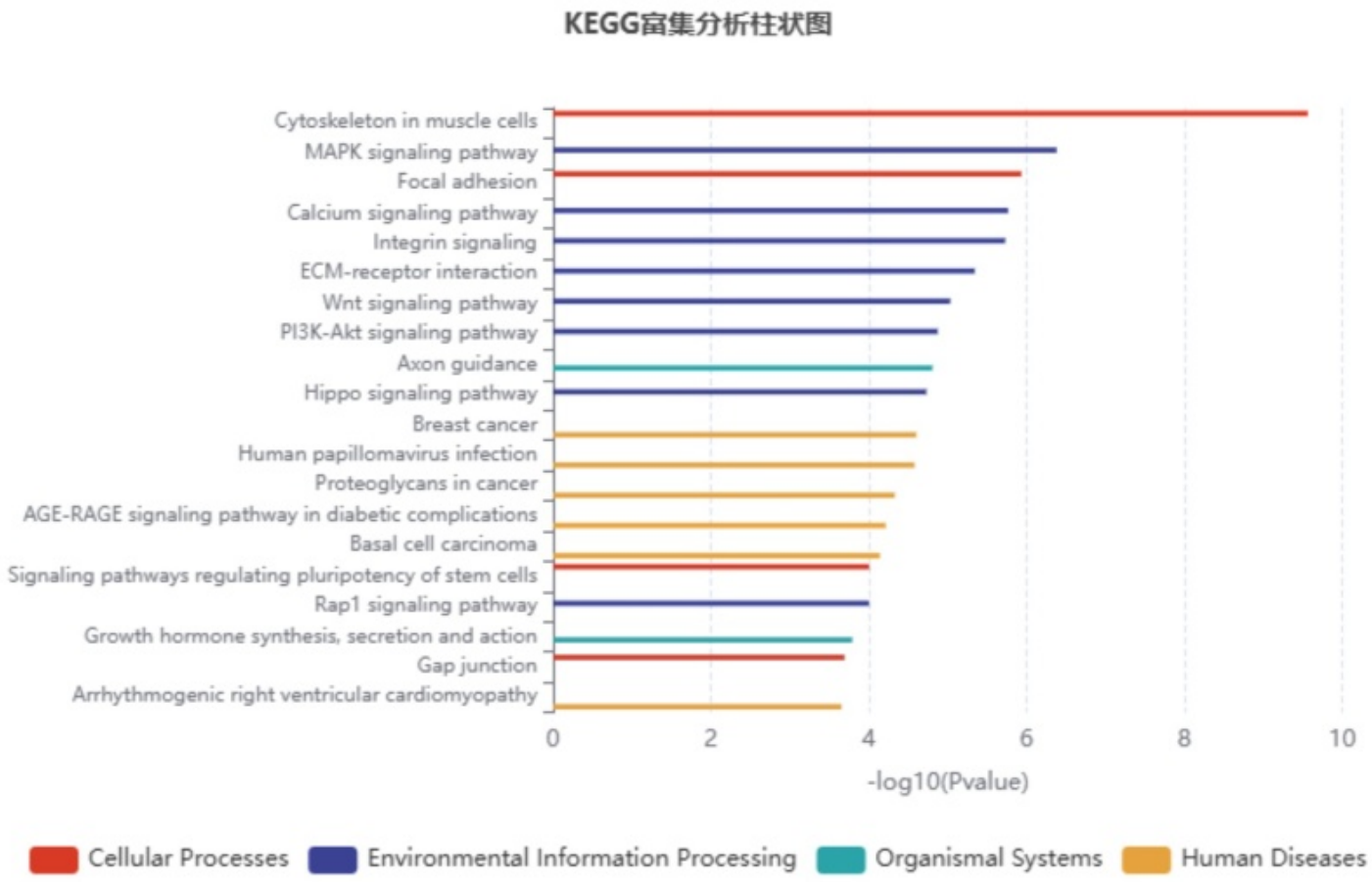
7.3GO富集分析气泡图



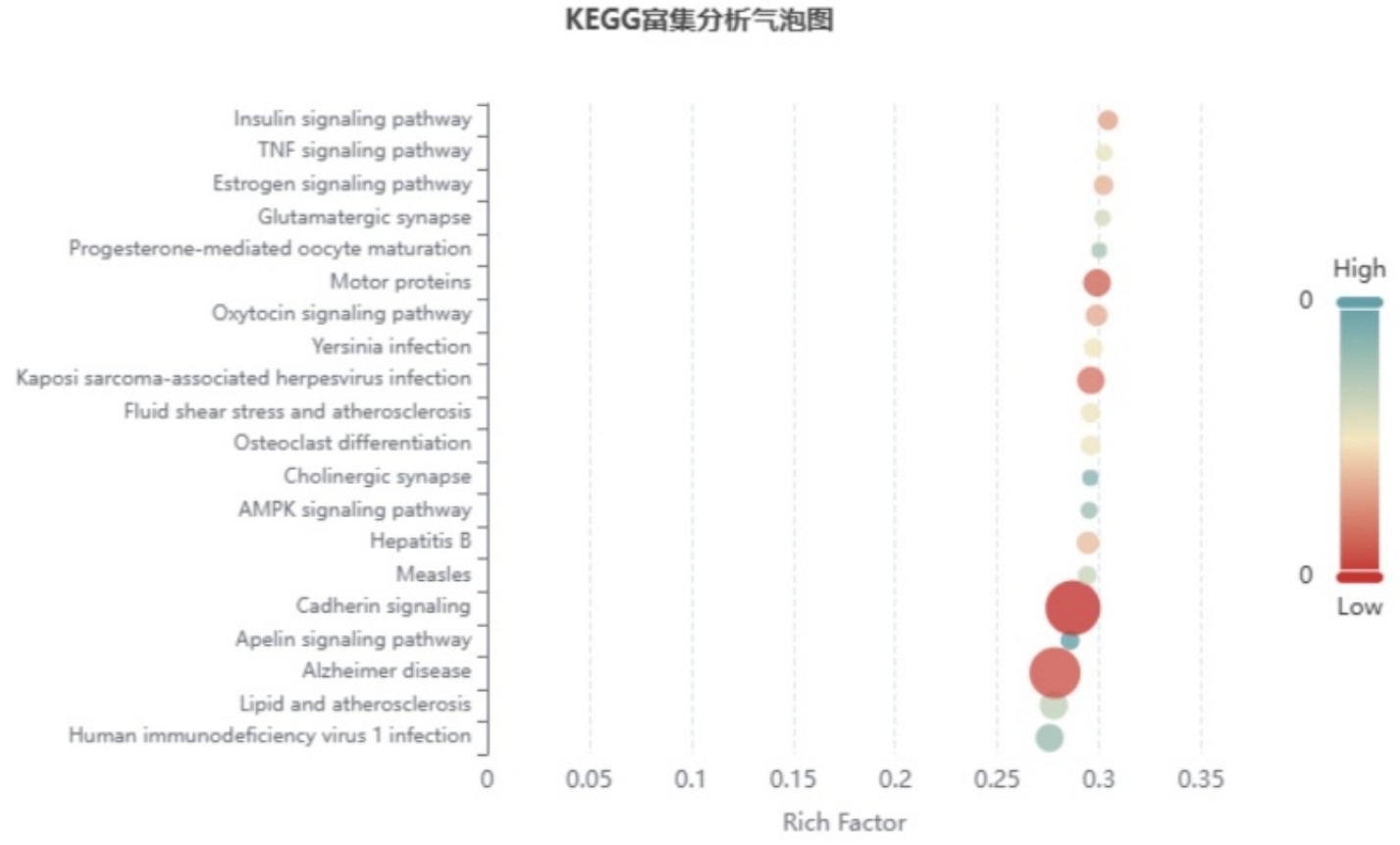
7.4KEGG富集分析

#	PathwayID	Pathway	level1	level2	Up	Down	
1	hsa04820	Cytoskeleton in muscle cells	Cellular Processes	Cell motility	38	56	
2	hsa04010	MAPK signaling pathway	Environmental Information Processing	Signal transduction	38	66	
3	hsa04510	Focal adhesion	Cellular Processes	Cellular community - eukaryotes	29	46	
4	hsa04020	Calcium signaling pathway	Environmental Information Processing	Signal transduction	28	61	
5	hsa04518	Integrin signaling	Environmental Information Processing	Signaling molecules and interaction	24	36	
6	hsa04512	ECM-receptor interaction	Environmental Information Processing	Signaling molecules and interaction	16	23	
7	hsa04310	Wnt signaling pathway	Environmental Information Processing	Signal transduction	21	44	
8	hsa04151	PI3K-Akt signaling pathway	Environmental Information Processing	Signal transduction	42	73	
9	hsa04360	Axon guidance	Organismal Systems	Development and regeneration	20	46	
10	hsa04390	Hippo signaling pathway	Environmental Information Processing	Signal transduction	22	36	

7.5KEGG富集分析柱状图



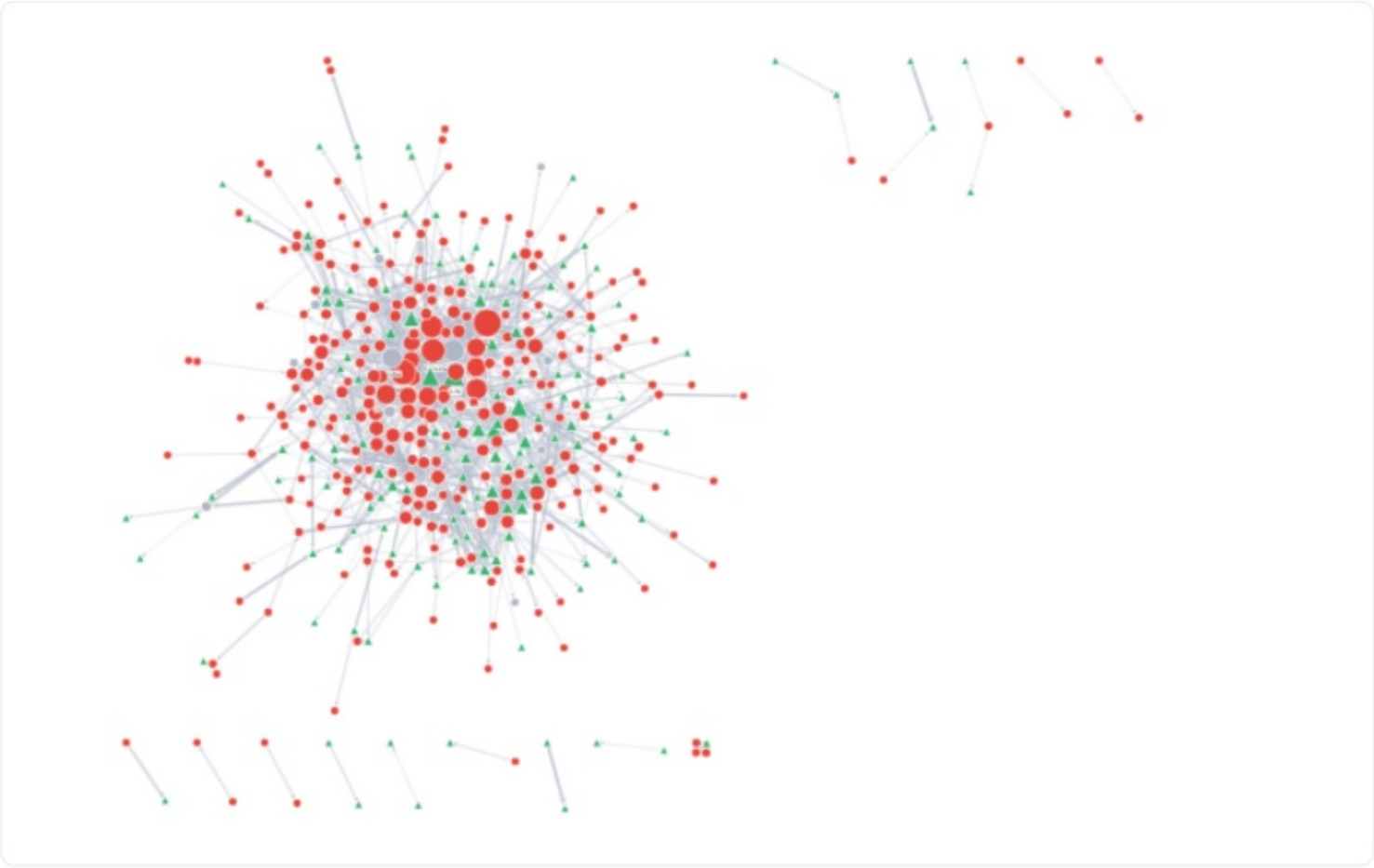
7.6KEGG富集分析气泡图



八、蛋白互作网络分析

构建差异表达基因的蛋白质-蛋白质相互作用网络。

8.1PPI网络图



8.2互作关系表

#	Node1	Node2	Score	Neighborhood	Gene Fusion	Cooccurrence	Coexpression	Experiments	Databases	Textmining
1	SLC7A2	SLC1A3	0.403	0	0	0	0.05	0.05	0	0.391
2	SLC7A2	SLC1A1	0.557	0	0	0	0.07	0.05	0	0.538
3	CD74	FGL2	0.404	0	0	0	0.257	0	0	0.231
4	CD74	HBD	0.424	0	0	0	0.047	0.095	0	0.386
5	CD74	CD34	0.428	0	0	0	0.08	0	0	0.403
6	CD74	JCHAIN	0.442	0	0	0	0.158	0	0	0.365
7	CD74	MCC	0.473	0	0	0	0	0.473	0	0
8	CD74	CXCL8	0.48	0	0	0	0.077	0	0	0.459
9	CD74	NME5	0.484	0.042	0	0	0.302	0.261	0	0.08
10	CD74	TM4SF20	0.491	0	0	0	0	0.473	0	0.074

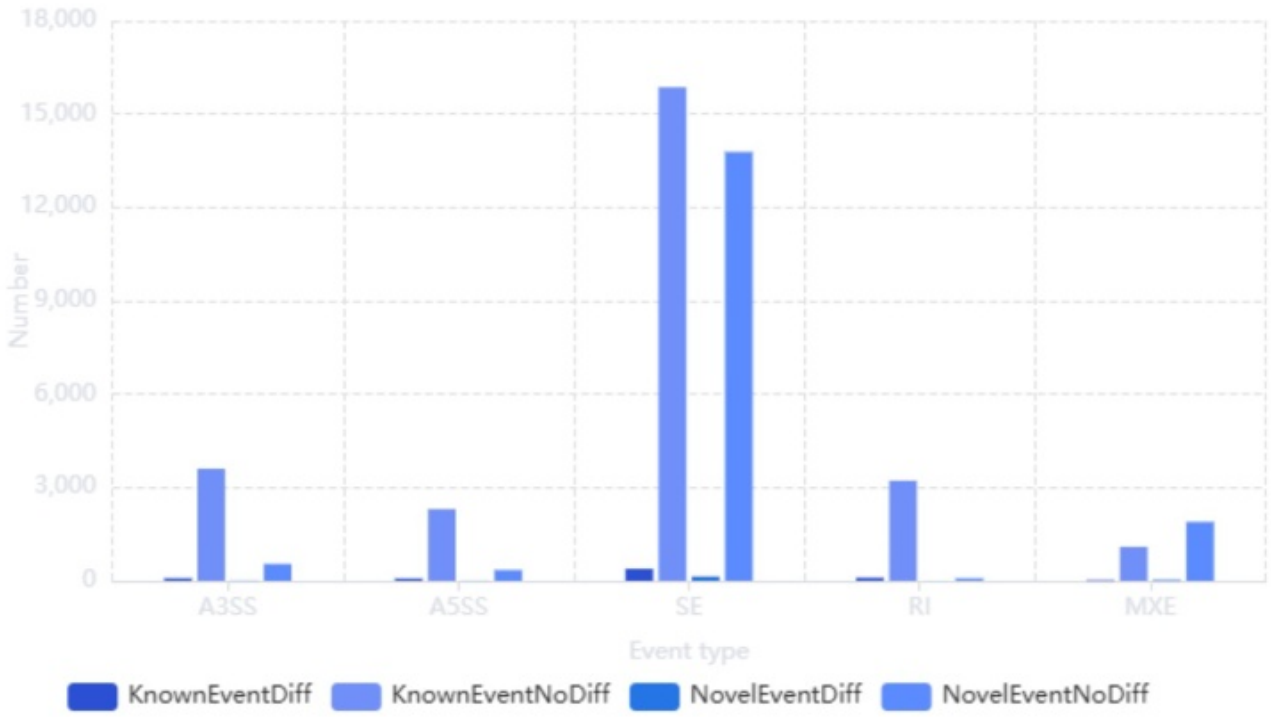
8.3节点属性表

#	Gene ID	Regulation	Type	Degree
1	SLC7A2	DOWN	mRNA	2
2	SLC1A3	DOWN	mRNA	15
3	SLC1A1	UP	mRNA	5
4	CD74	UP	mRNA	20
5	FGL2	UP	mRNA	7
6	HBD	UP	mRNA	4
7	CD34	UP	mRNA	37
8	JCHAIN		mRNA	4
9	MCC	DOWN	mRNA	1
10	CXCL8		mRNA	38

九、可变剪切分析

鉴定并分析不同类型的可变剪切事件。

9.1差异可变剪切分析图



9.2 A3SS可变剪切事件

#	EventID	GeneID	GeneName	Chr	Strand	Novel	coord1Start	coord1End	coord2Start	coord2End	coord3St
1	9	AGPAT3	NA	chr21	+	false	45387749	45387996	45387826	45387996	4538051
2	54	TRMU	NA	chr22	+	false	46746187	46746360	46746246	46746360	4674231
3	54	TRMU	NA	chr22	+	false	46746187	46746360	46746246	46746360	4674231
4	221	RPS9	NA	chr19	+	false	54710420	54711515	54711265	54711515	5471014
5	373	ZNF607	NA	chr19	-	false	38198814	38198913	38198814	38198910	3820059
6	373	ZNF607	NA	chr19	-	false	38198814	38198913	38198814	38198910	3820059
7	449	C19orf60	NA	chr19	+	false	18700222	18700493	18700288	18700493	1869980
8	449	C19orf60	NA	chr19	+	false	18700222	18700493	18700288	18700493	1869980
9	519	SMARCA4	NA	chr19	+	true	11094796	11095049	11094799	11095049	1107199
10	519	SMARCA4	NA	chr19	+	true	11094796	11095049	11094799	11095049	1107199

9.3 A5SS可变剪切事件

#	EventID	GeneID	GeneName	Chr	Strand	Novel	coord1Start	coord1End	coord2Start	coord2End	coord3Star
1	4	C21orf2	NA	chr21	-	false	45750345	45750799	45750705	45750799	45749773
2	21	HLCS	NA	chr21	-	false	38319632	38320587	38320465	38320587	38311131
3	24	EVA1C	NA	chr21	+	false	33873724	33873848	33873724	33873796	33876235
4	24	EVA1C	NA	chr21	+	false	33873724	33873848	33873724	33873796	33876235
5	122	ZNF415	NA	chr19	-	false	53636001	53636128	53636108	53636128	53625914
6	122	ZNF415	NA	chr19	-	false	53636001	53636128	53636108	53636128	53625914
7	150	FUZ	NA	chr19	-	false	50315703	50315993	50315871	50315993	50315507
8	150	FUZ	NA	chr19	-	false	50315703	50315993	50315871	50315993	50315507
9	239	ZNF493	NA	chr19	+	false	21579934	21580224	21579934	21580069	21587930
10	303	UBL5	NA	chr19	+	false	9938567	9938740	9938567	9938660	9939002

9.4 MXE可变剪切事件

#	EventID	GeneID	GeneName	Chr	Strand	Novel	coord1Start	coord1End	coord2Start	coord2End	coo
1	21	APP	NA	chr21	-	false	27369674	27369731	27372329	27372497	273
2	27	PDE9A	NA	chr21	+	false	44106301	44106372	44108026	44108104	440
3	125	ZNF562	NA	chr19	-	false	9768684	9768811	9770054	9770143	976
4	128	ZNF559	NA	chr19	+	true	9443690	9443789	9448471	9448541	943
5	129	ZNF559-ZNF177	NA	chr19	+	false	9443690	9443789	9448471	9448693	943
6	132	HSD11B1L	NA	chr19	+	false	5684829	5684916	5684975	5685130	568
7	133	HSD11B1L	NA	chr19	+	false	5684999	5685130	5686426	5686538	568
8	133	HSD11B1L	NA	chr19	+	false	5684999	5685130	5686426	5686538	568
9	134	HSD11B1L	NA	chr19	+	false	5685003	5685130	5686426	5686538	568
10	134	HSD11B1L	NA	chr19	+	false	5685003	5685130	5686426	5686538	568

9.5 RI可变剪切事件

#	EventID	GeneID	GeneName	Chr	Strand	Novel	coord1Start	coord1End	coord2Start	coord2End	coord3
1	13	U2AF1	NA	chr21	-	false	44513065	44514673	44513065	44513359	445145
2	19	HMGN1	NA	chr21	-	false	40717071	40720265	40717071	40717200	407202
3	45	LINC00899	NA	chr22	-	false	46438453	46439579	46438453	46438520	464394
4	45	LINC00899	NA	chr22	-	false	46438453	46439579	46438453	46438520	464394
5	85	MMP11	NA	chr22	+	false	24121373	24122689	24121373	24121555	241225
6	102	PI4KA	NA	chr22	-	false	21066980	21067678	21066980	21067085	210675
7	120	TPTEP1	NA	chr22	+	false	17094966	17096001	17094966	17095068	170955
8	133	ZNF446	NA	chr19	+	false	58991009	58991386	58991009	58991094	589912
9	136	ZNF154	NA	chr19	-	false	58208734	58212513	58208734	58209209	582123
10	138	ZNF667-AS1	NA	chr19	+	false	57005531	57006801	57005531	57005711	570064

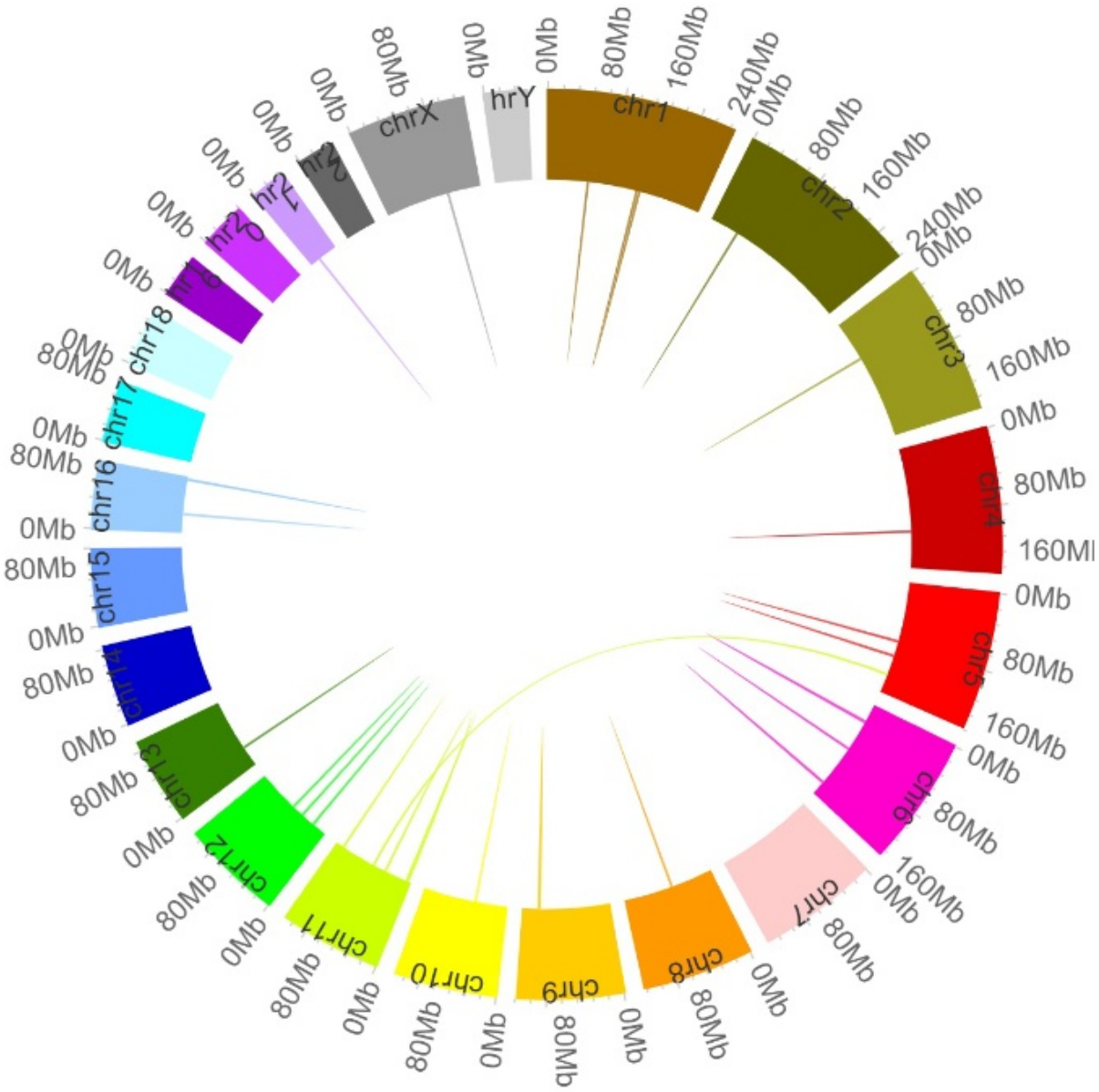
9.6 SE可变剪切事件

#	EventID	GeneID	GeneName	Chr	Strand	Novel	coord1Start	coord1End	coord2Start	coord2End	coord3St
1	105	SH3BGR	NA	chr21	+	false	40871748	40871841	40847060	40847141	4088092
2	105	SH3BGR	NA	chr21	+	false	40871748	40871841	40847060	40847141	4088092
3	285	NRIP1	NA	chr21	-	false	16386664	16386787	16333555	16340847	1641581
4	285	NRIP1	NA	chr21	-	false	16386664	16386787	16333555	16340847	1641581
5	323	DENND6B	NA	chr22	-	false	50754994	50755055	50754798	50754904	5075572
6	323	DENND6B	NA	chr22	-	false	50754994	50755055	50754798	50754904	5075572
7	324	DENND6B	NA	chr22	-	false	50754994	50755141	50754798	50754904	5075572
8	324	DENND6B	NA	chr22	-	false	50754994	50755141	50754798	50754904	5075572
9	660	THOC5	NA	chr22	-	false	29927066	29927099	29925150	29925228	2992781
10	660	THOC5	NA	chr22	-	false	29927066	29927099	29925150	29925228	2992781

十、融合基因分析

检测并注释可能的基因融合事件。

10.1融合基因圈图



10.2融合基因注释结果

#	SampleId	FusionName	JunctionReadCount	SpanningFragCount	est_J	est_S	SpliceType
1	SRR307902	TIMM23--PARGP1	50	29	24.81	14.5	ONLY_REF_SPLICE
2	SRR307902	HLA-S--HLA-A	9	0	9	0	ONLY_REF_SPLICE
3	SRR307902	KLC2--SPARC	8	0	8	0	INCL_NON_REF_SPLICE
4	SRR307902	RP11-14D22.2--PRICKLE2	5	1	5	1	INCL_NON_REF_SPLICE
5	SRR307902	PALM2--AKAP2	4	7	4	7	INCL_NON_REF_SPLICE
6	SRR307902	BOLA2B--RP11-345J4.8	3	15	1.5	7.5	INCL_NON_REF_SPLICE
7	SRR307902	RP11-444D3.1--SOX5	3	2	3	2	ONLY_REF_SPLICE
8	SRR307902	PRR16--CTD-2334D19.1	3	2	3	2	ONLY_REF_SPLICE
9	SRR307902	NRIP1--AF127936.7	3	4	3	2.4	ONLY_REF_SPLICE
10	SRR307902	NRIP1--AF127936.7	2	4	2	1.6	ONLY_REF_SPLICE